Government of Karnataka

DEPARTMENT OF SCHOOL EDUCATION (PRE-UNIVERSITY)

REVISED QUESTION BANK (2024 - 25)

For **Second Year P U C STATISTICS**

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UNIT – I VITAL STATISTICS

Choose the most appropriate answer from the choices given.

1. The statistical study of human population is called –
   a) Biology b) Sociology c) Demography d) Psychology

2. The formula for estimating the population between two census years is –
   a) $P_t = P_0+(B–D)+ (I–E)$ b) $P_t = P_0– (B+D) – (I+E)$
   c) $P_t = P_0+ (D+B) + (E+I)$ d) $P_t = P_0– (D–B) + (E–I)$

3. The births occurring to women of child bearing age is called
   a) Fertility b) Mortality c) Survival rate d) Fecundity

4. The capacity of a woman to bear children is called-
   a) Fertility b) Mortality c) Survival rate d) Fecundity

5. Generally the child bearing age for women is-
   a) 10–50 years b) 15–49 years c) 15–60 years d) 10–49 years

6. In a life table, generally the size of the cohort is –
   a) 100 b) 1,00,000 c) 1000 d) 10,000

7. The expected number of years that a new born baby would live is called- (Life expectancy of a new born baby)
   a) Longevity b) Cohort c) Radix d) Survival ratio

8. In a life table, the size of the cohort is
   a) Longevity b) Mortality ratio c) Radix d) Survival ratio

Fill in the blanks by choosing the appropriate word from those given in the brackets.
(Longevity, Fertility, Radix, Average, Demography, Annual, 15 to 49, Survive, Fecundity, Cohort)

1. The statistical study of human population is called _____.
2. The births occurring to women of child bearing age is called _____.
3. The capacity of a woman to bear children is called _____.
4. Generally the child bearing age for women is assumed in between _____ years.
5. TFR of a population in a year is the total of _____ age specific fertility rates of females of child bearing age.
6. TFR divided by 1000 gives _____ number of children born in life time to one woman.
7. TFR is based on the assumption that all women _____ throughout the child bearing age.
8. The expected number of years that a new born baby would live is called _____.
9. In a life table, the size of the cohort is called _____.

Match the following: (Direct answers are given)

	Infant		Child aged less than one year
	Births, deaths, migration etc.		Vital events
	Method of obtaining vital statistics		Census enumeration
	Deaths of new born babies within 28 days		Neonatal deaths
	Maternal mortality		Puerperal deaths
	Fertility		Births occurring to women of child bearing age
	Fecundity		The capacity of a woman to bear children
	15 – 49 years		Child bearing age for women
	N.R.R. per woman = 1		Population remains constant
	N.R.R. per woman = 0.95		Population is decreasing
	N.R.R. per woman = 1.2		Population is increasing
	Radix		1,00,000
	The size of the cohort		Radix
	Expected number of years that a new born baby would live		Longevity

One Mark Questions:

1. What is demography?
2. Mention a vital event occurring in the human population.
3. Mention a method of obtaining vital statistics.
4. Write the formula for estimating the population between two census years.
5. Mention a use of vital statistics.
6. Define fertility.
7. Define fecundity.
8. Mention a measure of fertility.
9. Define crude birth rate.
10. Write the formula of crude birth rate.
11. Mention a merit of crude birth rate.
12. Mention a demerit of crude birth rate.
13. Define general fertility rate.
14. Generally, what is the child bearing age (germination period) of women?
15. Write the formula of general fertility rate.
16. Mention a merit of general fertility rate.
17. Mention a demerit of general fertility rate.
18. Define age specific fertility rate.
19. Write the formula of age specific fertility rate.
20. Mention a merit of age specific fertility rate.
21. Mention a demerit of age specific fertility rate.
22. What is total fertility rate?
23. Mention a merit of total fertility rate.
24. Mention a demerit of total fertility rate.

25. Mention a measure of reproduction rate.
26. Define gross reproduction rate.
27. If GRR is 1300, then find number of female births to a woman, if she is alive till the end of reproduction age.
28. Define net reproduction rate.
29. In a situation, if GRR per woman is greater than 1 and NRR per woman is less than 1 then what is your conclusion about population?
30. If NRR per woman is less than 1, then what it indicates about the population?
31. If NRR per woman is one, then what it indicates about the population?
32. If NRR per woman is more than 1, then what it indicates about the population?
33. Define mortality.
34. Mention a measure of mortality.
35. Define crude death rate.
36. Write the formula of crude death rate.
37. Mention a merit of crude death rate.
38. Mention a demerit of crude death rate.
39. Define age specific death rate.
40. Write the formula of age specific death rate.
41. Write a merit of age specific death rate.
42. Write a demerit of age specific death rate.
43. Define standardized death rate.
44. Which death rate is used to compare the health condition of two populations?
45. Write a merit of standardized death rate.
46. Write a demerit of standardized death rate.
47. Define infant mortality rate.
48. Write the formula of infant mortality rate.
49. Define neo-natal mortality rate.
50. Write the formula of neo-natal mortality rate.
51. Define maternal mortality rate.
52. Write the formula of maternal mortality rate.
53. Define life table.
54. Define cohort.
55. Define radix.
56. What is longevity?
57. Mention a use of life table.
58. Write the formula of calculating dx in life table.
59. Define mortality ratio.
60. Write the formula of mortality ratio.
61. Define survival ratio.
62. Write the formula of survival ratio.
63. Define expectation of life.
64. Write the formula of expectation of life.

Five Mark Questions:

1. For the following data compute general fertility rate.

Age [in years]	Male Population	Female Population	Number of live births
0 - 14	11,000	10,000	-
15 - 19	9,000	8,000	320
20 - 24	12,000	11,000	660
25 - 29	16,000	15,000	1,350
30 - 34	21,000	20,000	2,000
35 - 39	15,000	15,000	1,035
40 - 44	7,000	6,000	180
45 - 49	4,000	5,000	55
50 & above	6,000	9,000	-

2. Calculate CBR, GFR and ASFR for the age group [15-19].

Age [in years]	Male Population	Female Population	Number of live births
0 - 9	6400	5197	0
10 - 14	5430	6153	0
15 - 19	6300	7888	510
20 - 24	2300	3444	880
25 - 39	4700	3800	277
40 - 49	5600	4400	45
50 & above	2800	1119	0

3. For the following data, find TFR.

Age group [in years]	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44	45 - 49
Female population	8,000	11,000	15,000	20,000	15,000	6,000	5,000
Live births	320	660	1,350	2,000	1,035	180	55

4. Find total fertility rate for the following data.

Age group [in years]	Female population	Live births
15 - 19	14000	840
20 - 24	15000	1350
25 - 29	14000	2660
30 - 34	12000	1200
35 - 39	13000	1040
40 - 44	12000	960
45 - 49	10000	500

5. For the following data, calculate gross reproduction rate.

Age group [in years]	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44	45 - 49
Female population	14,000	15,000	14,000	12,000	13,000	12,000	10,000
Female births	630	870	980	660	650	240	30

6. Find the gross reproduction rate from the following data. Obtain the average number of female children born to woman of child bearing age.

Age [in years]	Female Population	Female births
15 - 19	10000	200
20 - 24	9000	540
25 - 29	8000	400
30 - 34	7000	280
35 - 39	6000	180
40 - 44	5000	100
45 - 49	4000	40

7. For the following data, calculate net reproduction rate.

Age group [in years]	15-19	20 24	25 - 29	30-34	35-39	40-44	45-49
Female population	8,000	9,000	10,000	9,000	8,000	7,000	6,000
Female births	240	450	600	360	160	70	30
Survival ratio	0.90	0.88	0.85	0.84	0.82	0.80	0.80

8. Compute age specific death rates for the following data

Age [in years]	Population	Deaths
Under 10	12,000	450
10 - 19	20,000	480
20 – 39	40,000	800
40 – 59	20,000	670
60 & above	8,000	500

9. Compute standardized death rates for towns A and B. State which town is healthier.

Age [in years]	Death rates		Standard Population
	Town A	Town B	
0 – 9	18	20	15,000
10 – 29	10	9	35,000
30 – 59	15	8	30,000
60 & above	20	24	20,000

Ten Mark Questions:

1. For the following data, compute CBR, GFR and TFR.

Age [in years]	Male Population	Female Population	Number of live births
0 - 14	11,000	10,000	-
15 - 19	9,000	8,000	320
20 - 24	12,000	11,000	660
25 - 29	16,000	15,000	1,350
30 - 34	21,000	20,000	2,000
35 - 39	15,000	15,000	1,035
40 - 44	7,000	6,000	180
45 - 49	4,000	5,000	55
50 & above	6,000	9,000	-

2. For the following data, compute the GRR, NRR and hence comment on the results.

Age group [in years]	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44	45 - 49
Female population	8,000	9,000	10,000	9,000	8,000	7,000	6,000
Female births	240	450	600	360	160	70	30
Survival ratio	0.90	0.88	0.85	0.84	0.82	0.80	0.80

3. For the following data, compute the GRR, NRR and hence comment on the results.

Age group [in years]	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44	45 - 49
Female population	3,000	2,500	2,200	2,000	1,800	1,500	1,200
Female births	60	100	132	80	54	30	12
Survival ratio	0.9	0.9	0.8	0.8	0.8	0.7	0.7

4. For the following data, compute the GRR and NRR. On the basis of NRR draw inference about the population.

Age group [in years]	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44	45 - 49
Female population	15000	11000	16000	17000	16000	15000	10000
Female births	180	715	960	680	352	120	10
Survival ratio	0.95	0.92	0.89	0.87	0.85	0.83	0.80

5. From the following data, compute standardized death rates for Town A and Town B. Comment on the results.

Age group [in years]	Town - A		Town - B		Standard Population
	Population	Deaths	Population	Deaths	
0 – 20	5,000	100	7,000	105	4,000
20 – 50	14,000	392	15,000	465	16,000
50 – 70	20,000	300	25,000	500	18,000
70 & above	1,000	200	3,000	390	2,000

6. From the following data, calculate the STDRs for locality A and locality B. Comment on the results.

Age group [in years]	Locality A		Locality B		Standard Population
	Population	Deaths	Population	Deaths	
0 – 20	4,000	68	8,000	160	6,000
20 – 40	9,000	54	13,000	65	12,000
40 – 60	7,000	91	10,000	130	8,000
60 & above	3,000	129	4,000	160	4,000

7. From the following data, compute standardized death rates for village A and village B. Which village is healthier?

Age group [in years]	Village - A		Village - B		Standard Population
	Population	Deaths	Population	Deaths	
0 – 20	4,000	36	3,000	30	2,000
20 – 40	12,000	48	20,000	100	3,000
40 – 60	6,000	60	4,000	48	6,000
60 & above	8,000	152	3,000	60	4,000

8. From the following data, compute standardized death rates by taking locality B population as standard and comment on results.

Age group [in years]	Locality A		Locality B	
	Population	Deaths	Population	Deaths
0 – 20	4,000	68	6,000	120
20 – 40	9,000	54	12,000	60
40 – 60	7,000	91	8,000	104
60 & Above	3,000	129	4,000	160

9. From the following data, compute standardized death rates and comment.

Age group [in years]	Locality A		Locality B [Standard]	
	Population	Deaths	Population	Deaths
0 – 20	3,000	45	4,000	40
20 – 40	8,000	48	10,000	60
40 – 60	7,000	42	10,000	70
60 & Above	5,000	110	6,000	108

10. From the following data, compute standardized death rates and comment.

Age group [in years]	Locality A [Standard]		Locality B	
	Population	Deaths	Population	Deaths
Below 10	5,000	140	5,000	145
10 – 25	12,000	50	14,000	60
25 – 65	15,000	80	20,000	90
65 & above	4,000	150	1,000	110

UNIT – II INDEX NUMBERS

Choose the most appropriate answer from the choices given.

1. The value of the index number for the base year is –
   a) 50 b) 100 c) 150 d) 200

2. If the general price level goes up by 80% between 2020 and 2022, the index number for 2022 with base 2020 is-
   a) 20 b) 80 c) 100 d) 180

3. The prices of items increased by 10% in 2012 as compared to 2010. The index number for the year 2010 is,
   a) 110 b) 10 c) 100 d) 0

4. The prices of items decreased by 10% in 2020 as compared to 2018. The index number for the year 2020 is,
   a) 110 b) 10 c) 100 d) 90

5. If price during the current year is double the price during the base year, then the price relative is
   a) 50 b) 100 c) 150 d) 200

6. The weights used in the construction of Laspeyre's price index number are –
   a) Base year prices b) Current year prices
   c) Base year quantities d) Current year quantities

7. The weights used in the construction of Paasche's price index number are –
   a) Base year prices b) Current year prices
   c) Base year quantities d) Current year quantities

8. The weights used in construction of Marshall-Edgeworth's price index numbers
   a) q_0 b) q_1 c) $\left(\frac{q_0+q_1}{2}\right)$ d) $(q_0 \div q_1)$

9. The condition for the 'time reversal test' to be satisfied with usual notation is –
   a) $P_{01} \times P_{10} = 1$ b) $P_{01} \times P_{10} = 0$ c) $P_{01} \times Q_{01} = V_{01}$ d) $P_{01} \times V_{01} = Q_{01}$

10. The condition for the 'factor reversal test' to be satisfied with usual notation is –
    a) $P_{01} \times P_{10} = 1$ b) $P_{01} \times P_{10} = 0$ c) $P_{01} \times Q_{01} = V_{01}$ d) $P_{01} \times V_{01} = Q_{01}$

11. Factor reversal test is satisfied by
    a) Laspeyre's index number b) Marshall – Edgeworth's index number
    c) Fisher's index number d) Kelly's index number

12. Laspeyre's's index number satisfies
    a) Unit test b) Circular test c) Time reversal test d) Factor reversal test

13. Paasche's index number satisfies
    a) Unit test b) Circular test c) Time reversal test d) Factor reversal test

14. Dorbish-Bowley's index number satisfies
    a) Unit test b) Circular test c) Time reversal test d) Factor reversal test

Fill in the blanks by choosing the appropriate word from those given in the brackets.
(Laspeyre's, Paasche's, Fisher's, Retail, Economic barometers, 100, Averages, Base year, Current year, q_0, Simple aggregative price, q_1, Price, quality, Geometric mean)

1. Index numbers are known as _____.
2. Index numbers are specialized type of _____.
3. The value of the index number for the base year is _____.
4. In the computation of index numbers, most preferred type of average is _____.
5. The weight that is used in the construction of Laspeyre's price index number is _____.
6. The weight that is used in the construction of Paasche's price index number is _____.
7. Laspeyre's quantity index number uses the _____ prices as weights.
8. Paasche's quantity index number uses the _____ prices as weights.
9. _____ index number shows an upward bias.
10. _____index number shows downward bias.
11. _____ index number satisfies both TRT and FRT.
12. _____ index number does not satisfy unit test.
13. The _____ price of the commodities used in the construction of cost of living index number.
14. A value weight means quantity produced multiplied by the _____ of the commodities.
15. In constructing index number, the _____ of the product is not taken into consideration.

Match the following: (Direct answers are given)

	Index numbers		Economic barometers
	Base period		Economically stable
	P_{01} = 80		20% of the price of the commodity decreases
	P_{01} = 140		40% of the price of the commodity increases
	Laspeyre's index number		Upward bias
	Paasche's index number		Downward bias
	Fisher's index number		Free of bias

	Fisher's index number		Satisfies both TRT and FRT
	Fisher's index number		Ideal
	Dorbish-Bowley's index number		Free of bias
	Index number which doesn't satisfy unit test		Simple aggregative price index number
	$P_{01} \times P_{10} = 1$		Time reversal test
	$P_{01} \times Q_{01} = V_{01}$		Factor reversal test
	$P_{01} \times P_{12} \times P_{20} = 1$		Circular test
	Current year weight		Paasche's index number
	Base year weight		Laspeyre's index number
	Retail price of commodities		Cost of living index number

One Mark Questions:

1. In index number what is meant by base year?
2. In index number what is meant by current year?
3. Define price relative.
4. Write the formula of price relative.
5. Write the formula of quantity relative.
6. What is the value of index number for the base year?
7. If the price during the current year is triple the price during the base year, what is the value of price relative?
8. If the price relative is 175, what would you conclude?
9. If the general price level goes up by 80% between 2000 and 2012, what is the index number for 2012 with base 2000?
10. If the quantity index number for current year is 80, then what would you conclude?
11. Mention a characteristic of index numbers.
12. Mention a use of index numbers.
13. Mention a limitation of index numbers.
14. Define price index number.
15. Mention two steps involved in the construction of index number.
16. Name the common average used in the construction of index numbers.
17. Which average is considered as the best average in the construction of index number?
18. Why GM is considered as the best average in the construction of index number?
19. What is simple aggregative price index number?
20. Write down the formula of simple aggregative price index number.
21. Write down the formula of simple arithmetic mean price index number.
22. Write down the formula of simple geometric mean price index number.
23. Write down the formula of weighted arithmetic mean price index number.
24. Write down the formula of weighted geometric mean price index number.
25. Write down the formula of Laspeyre's price index number.
26. Which weight is used in the construction of Laspeyre's price index number?
27. Write down the formula of Paasche's price index number.
28. Which weight is used in the construction of Paasche's price index number?
29. Write down the formula of Marshall-Edgeworth's price index number.

30. Which system of weight is used in the construction of Marshall-Edgeworth's price index number?
31. State the relation between Laspeyre's, Paasche's and Dorbish–Bowley's index numbers.
32. Write down the formula of Dorbish – Bowley's price index number.
33. State the relation between Laspeyer's, Paasche's and and Fisher's index numbers.
34. Write down the formula of Fisher's price index number.
35. Write down the formula for Kelly's fixed weight price index number.
36. Which weight is used in the construction of Kelly's price index number?
37. Write down the formula of Laspeyre's quantity index number.
38. Which weight is used in the construction of Laspeyre's quantity index number?
39. Write down the formula of Paasche's quantity index number.
40. Which weight is used in the construction of Paasche's quantity index number?
41. Write down the formula of Marshall-Edgeworth's quantity index number.
42. Which system of weight is used in the construction of Marshall-Edgeworth's quantity index number?
43. Write down the formula of Dorbish – Bowley's quantity index number.
44. Write down the formula of Fisher's quantity index number.
45. Write down the formula of value index number.
46. What do you mean by unit test?
47. Name the index number which does not satisfy unit test.
48. State the condition required to satisfy Time Reversal Test (TRT).
49. Name the index number which satisfies TRT.
50. Does Marshall - Edgeworth's index number satisfies TRT?
51. State the condition required to satisfy Factor Reversal Test (FRT).
52. Name the index number which satisfies FRT.
53. Does Marshall - Edgeworth's index number satisfies FRT?
54. Name the index number which satisfies both TRT and FRT.
55. State the condition required to satisfy circular test.
56. Name the index number which satisfies circular test.
57. Which index number shows upward bias?
58. Why Laspeyre's price index number shows upward bias?
59. Which index number shows downward bias?
60. Why Paasche's price index number shows downward bias?
61. Is Marshall - Edgeworth's index number free from bias?
62. Why Fisher's index number is free from bias?
63. Define consumer price index number (cost of living index number).
64. Write a use of consumer price index number.
65. Which price of the commodities is used in the construction of cost of living index number?
66. State a method used to compute consumer price index number.
67. Write the formula for computing CPI by aggregative expenditure method.
68. Write the formula for computing CPI by family budget method.

Five Mark Questions:

1. Following are the prices (in Rs.) of items in 2010 and 2015. Calculate simple arithmetic mean price index number. Comment on the result.

Item		A	B	C	D	E	F
Price (Rs.)	2010	50	60	20	50	80	125
	2015	55	75	30	75	90	130

2. The following are the prices (in Rs.) of items in 2010 and 2015. Find simple geometric mean price index number.

Item		A	B	C	D	E	F
Price (Rs.)	2010	50	60	20	50	80	125
	2015	55	75	30	75	90	130

3. Calculate simple geometric mean price index number for the following data.

Item		A	B	C	D	E
Price (Rs.)	Base year	24	40	20	16	50
	Current year	30	35	24	16	60

4. For the following data calculate the weighted arithmetic mean price index number.

Commodity		Wheat	Gram	Rice	Pulses
Price (Rs.)	2005	50	60	20	50
	2010	55	75	30	75
Weight		4	2	3	1

5. Find the weighted A.M price index number from the following data.

Item		A	B	C	D	E
Weight in %		25	10	20	15	30
Price (Rs.)	2000	120	30	50	25	40
	2005	100	30	40	20	50

6. Find weighted G.M. price index number from the following data.

Item		A	B	C	D	E
Weight in %		30	15	20	10	25
Price (Rs.)	Base year	100	20	70	20	40
	Current year	90	20	60	15	55

7. Calculate the weighted geometric mean price index number for the following data.

Item		Stereo	Television	Radio
Price Rs/ Unit	Base year	20,000	15,000	500
	Current year	25,000	20,000	800
Weight		30	50	20

8. By using the following data compute suitable index number and comment on the result.

Item	Base year Quantity	Price (Rs.)	
		Base year	Current year
A	20	8	4
B	40	12	10
C	40	20	15
D	50	40	25
E	50	50	10

9. Compute suitable index number from the following data and comment on the result.

Item	p_0	p_1	q_1
I	5	6	30
II	3	4	15
III	4	5	18
IV	2	2	10

10. Compute Kelly's price index numbers for 2005 from the following data. Comment on the result.

Item	Price (Rs.)		Quantity
	2000	2005	
A	15	22	15.5
B	20	27	12.5
C	4	7	7.5
D	10	20	7.5

11. Calculate Kelly's price index number from the following data. Comment on the result.

Item	No. of units sold	Price (Rs.)	
		Base year	Current year
A	22	25	30
B	40	60	75
C	35	100	110
D	45	70	90

12. Compute Kelly's price index number for the following data. Comment on the result.

Item	Price (Rs.)		Quantity of consumption
	2010	2012	
A	10	12	20
B	16	18	15
C	9	10	10
D	11	14	25

13. Compute Kelly's price index number for the following data. Comment on the result.

Item	A	B	C	D
p_0	14	22	10	8
p_1	15	24	12	10
q	5	4	10	12

14. Compute suitable index number from the following data. Comment on the result.

Item	Unit	Quantity		Price in 1990
		1990	1995	
A	kg	150	160	10
B	kg	90	100	12
C	meter	60	60	15
D	packets	50	40	9

15. Compute suitable index number from the following data. Comment on the result.

Item	p_0	q_0	q_1
I	30	6	5
II	15	4	3
III	18	5	4
IV	10	2	2

16. Calculate suitable index number from the following data. Comment on the result.

Item	Current year price	Quantity	
		Base year	Current year
A	30	8	10
B	45	10	15
C	100	7	10
D	22	20	25

17. For the following data show that Marshall-Edgeworth's price index number satisfies TRT.

Item	Base year		Current year	
	Price	Quantity	Price	Quantity
A	4	4	6	2
B	6	4	4	8
C	8	5	10	3

18. From the following data compute value index number for the year 2010 on the basis of 2008.

Item	2008		2010	
	Price (in Rs.)	Quantity	Price (in Rs.)	Quantity
A	9	10	10	11
B	10	9	11	10
C	7	8	8	10
D	15	8	15	9

19. For the following data compute value index number for the current year. Comment on the result.

Item	Base year		Current year	
	Price	Quantity	Price	Quantity
A	5	25	6	30
B	10	5	15	4
C	3	40	2	50
D	6	30	8	35

20. For the following data find the consumer price index number for the year 2012 with respect to the base year 2005 by aggregative expenditure method.

Commodity	Unit	No. of units (quantities) in 2005	Price (in Rs. per unit)	
			2005	2012
Rice	quintal	2	1600	3800
Dhal	quintal	0.2	2100	6400
Sugar	kg	30	15	32
Tea	kg	3	60	100
Miscellaneous	monthly	12	2000	3000

21. Calculate the cost of living index number by aggregative expenditure method.

Commodity	Base year		Current year price (in Rs.)
	Price (in Rs.)	Quantity	
Wheat	26	40	30
Pulses	48	5	60
Salt	2	4	2.5
Oil	150	15	170
Others	1000	6	1400

22. For the following data find consumer price index number by aggregative expenditure method.

Item	No. of units (quantities) in 2010	Price per unit in Rs.	
		2010	2015
A	100	8	12
B	25	6	7
C	10	5	5
D	20	48	52
E	25	15	16
F	30	9	27

23. For the following data calculate the cost of living index number by aggregative expenditure method.

Commodity	Base year		Current year price (in Rs.)
	Price (in Rs.)	Expenditure (in Rs.)	
Rice	12	960	17
Sugar	24	360	30
Tea	200	400	300
Pulses	40	200	50
Fuel	500	4000	600
Others	1000	20000	2500

24. For the following data calculate the cost of living index number by family budget method.

Group	Weight	Price (in Rs.)	
		Base year	Current year
Food	10	2000	2500
Housing	5	800	1200
Clothing	3	400	500
Fuel	7	500	700
Miscellaneous	5	800	1000

25. For the following data calculate the cost of living index number by family budget method.

Group	Price (in Rs.)		Weight
	Base year	Current year	
Food	130	170	30
Clothing	50	60	12
Fuel	90	110	8
Entertainment	30	50	15
Medicine and Education	40	70	10
Others	50	90	15

26. For the following data calculate the cost of living index number.

Group	Price (in Rs.)		Weight
	Base year	Current year	
Food	2000	2500	10
Clothing	800	1200	4
Housing	2500	3000	12
Fuel	500	400	8
Miscellaneous	800	1000	6

Ten mark questions:

1. Find Laspeyre's, Paasche's and Fisher's price index numbers for 2000 from the following data.

Item	1995		2000	
	Price (Rs.)	Quantity	Price (Rs.)	Quantity
A	6	50	10	56
B	2	100	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36

2. From the following data compute Marshall-Edgeworth's and Dorbish-Bowley's price Index numbers.

Item	Price (in Rs.)		Quantity	
	Base year	Current year	Base year	Current year
A	6	10	50	56
B	2	2	100	120
C	4	6	60	60
D	10	12	30	24
E	8	12	40	36

3. Find Laspeyre's, Paasche's and Dorbish-Bowley's price index numbers for the following data.

Item	2004		2008	
	Price (Rs.)	Quantity	Price (Rs.)	Quantity
A	10	5	12	4
B	15	8	18	7
C	6	3	4	5
D	3	4	3	5

4. From the following data compute Marshall-Edgeworth's and Fisher's price Index numbers.

Item	Price (in Rs.)		Quantity	
	Base year	Current year	Base year	Current year
A	10	12	60	60
B	4	5	100	90
C	5	6	70	80
D	6	6	60	40

5. From the following data find Laspeyre's, Paasche's and Marshall-Edgeworth's price Index numbers.

Item	Base year		Current year	
	Price (Rs.)	Expenditure (Rs.)	Price (Rs.)	Expenditure (Rs.)
A	5	25	10	60
B	1	10	2	24
C	4	14	8	40
D	2	40	5	75

6. From the following data compute Dorbish-Bowley's and Fisher's price Index numbers.

Item	Price (Rs.)		Expenditure (Rs.)	
	Base year	Current year	Base year	Current year
1	2	5	50	60
2	4	8	20	48
3	1	2	8	20
4	5	10	30	70

7. For the following data show that Fisher's index number satisfies both time reversal and factor reversal tests.

Item	2004		2006	
	Price (Rs.)	Quantity	Price (Rs.)	Quantity
A	8	15	9	15
B	7	12	8	13
C	10	10	10	10
D	12	14	15	16

8. For the following data verify whether Fisher's index number satisfies TRT and FRT.

Item	Price (Rs.)		Quantity	
	Base year	Current year	Base year	Current year
A	4	6	4	2
B	6	4	4	8
C	8	10	5	3

9. For the following data verify whether Marshall-Edgewroth's index number satisfies TRT and FRT.

Item	Base year		Current year	
	Price (Rs.)	Quantity	Price (Rs.)	Quantity
A	4	4	6	2
B	6	4	4	8
C	8	5	10	3

10. Using the following data verify whether
    a. Laspeyre's index number satisfies FRT
    b. Marshall-Edgewroth's price index number satisfies TRT.

Commodity	Base year		Current year	
	Price (Rs.)	Quantity	Price (Rs.)	Quantity
Rice	40	20	45	22
Wheat	25	16	30	15
Oil	95	8	95	9
Fish	110	10	120	10
Milk	20	6	30	7

UNIT – III TIME SERIES ANALYSIS

Choose the most appropriate answer from the choices given.

1. A graphic representation of time series data is called
   a) Frequency polygon b) Ogive
   c) Frequency curve d) Historigram

2. The general tendency of the time series data to increase or to decrease or to remain constant during a long period of time is called
   a) Secular trend b) Seasonal variation
   c) Cyclical variation d) irregular variation

3. The regular, periodic and short term variation in a time series is called
   a) Secular trend b) Seasonal variation
   c) Cyclical variation d) irregular variation

4. The sudden unexpected variation in a time series is called
   a) Secular trend b) Seasonal variation
   c) Cyclical variation d) irregular variation

5. The method in which the original time series data is divided into two equal parts and averages are calculated for both the parts is known as
   a) Graphical method b) semi averages method
   c) Moving averages method d) least squares method

Fill in the blanks by choosing the appropriate word from those given in the brackets.
(Semi-averages, Indices, Normal, Irregular, An upward, downward, Historigram, Secular trend, Seasonal, Cyclical)

1. A graphic representation of time series data is called _____.
2. The general tendency of the time series data to increase or to decrease or to remain constant during a long period of time is called _____.
3. Increase in the price of gold in the past many years is _____ trend.
4. Decrease in the death rates in the past many years is a _____ trend.
5. The regular, periodic and short term variation in a time series is called _____ variation.
6. Seasonal _____ are used for the measurement of seasonal variation.
7. The sudden unexpected variation in the time series is called _____ variation.
8. In the _____ method the original time series data is divided into two equal parts and averages are calculated for both the parts.
9. The process of minimisation of sum of squared errors results in some equations which are called _____ equations.

Match the following. (Direct answers are given)

	Time series		Data arranged chronologically
	Graphic representation of time series		Historigram
	Period of cyclical variation		Within one year
	Business cycle		Oscillatory variation
	Irregular variation		Unpredictable
	$y = a + bx$		Straight line trend
	$y = a + bx + cx^2$		Parabolic (quadratic) trend
	$y = ab^x$		Exponential (Logarithmic) trend

One mark questions:

1. Define time series.
2. Give an example for time series.
3. What is Historigram?
4. Mention a use of time series.
5. Mention two components of time series.
6. Define secular trend.
7. Give an example for secular trend.
8. Give an example for upward trend.
9. Give an example for downward trend.
10. Define seasonal variation.
11. Is seasonal variation of the time series predictable?
12. Mention a factor causing seasonal variation.
13. Give an example for seasonal variation.
14. Which component of a time series is associated with 'Increase in the sales of text books in the month of June'?
15. Which index is used for the measurement of seasonal variation?
16. Define cyclical variation.
17. Define irregular (random) variation.
18. Mention a factor causing random variation.
19. Give an example for random variation.
20. Which component of a time series is associated with 'Sudden increase in theprice of vegetables due to strike'?
21. Which variation of the time series is unpredictable?
22. Name two methods of measuring trend.
23. Write a merit of method moving averages for measuring trend.
24. Write a demerit of method moving averages for measuring trend.
25. Write a merit of method least squares for measuring trend.

Two mark questions:

1. Define time series. Give an example.
2. Mention two uses of time series.
3. Mention four components of time series.
4. Define secular trend. Give an example.
5. Define seasonal variation. Give an example.
6. What are the different phases in a business cycle?
7. Diagrammatically represent 'Business Cycle' with stages.
8. Define random variation. Give an example.
9. Mention a factor causing following variations:
   i) Seasonal variation ii) Irregular variation

10. Which component of a time series is associated with the following sentences?
    (a) Fall in death rate due to advance in science.
    (b) An increase in employment during harvest season.
11. Name four methods of measuring trend.
12. State two conditions of least squares method of measuring trend.
13. Write down the normal equations for fitting linear trend.
14. Given a second degree trend equation $Y = 25 - 0.75X + 2X^2$. Estimate the value of Y, when X = 3.

Five mark questions:

1. Obtain trend values by 3 yearly moving averages for the following data.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
No. of Students	15	18	17	20	23	25	29	36	33	40

2. The following figures relate to the profits of 10 years. Find the trend values of the profits by 3 yearly moving averages method.

Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Profits (in lakhs)	30	27	39	36	42	48	45	51	48	54

3. Find trend values by five yearly moving averages for the following time series.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008
Value	10	15	18	21	25	30	33	40	50

4. For the following time series compute trend values by five yearly moving averages.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Value	14.6	15.9	17.4	17.9	16.5	16	17.1	18.3	19.6	20.1	18.5

5. Calculate trend values for the following data by 5 yearly moving averages.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Sales ('000)	18	18	21	20	16	17	17	16	21	20	20

6. Compute trend values by four yearly moving averages for the following data.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
Sales ('000)	75	60	55	60	65	70	70	75	85	70

7. Estimate the trend values by taking four yearly moving averages.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
Value	12	25	39	54	70	87	105	100	82	65

8. Calculate trend values by 4 yearly moving averages for the following data.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
Profit (in crores)	80	85	82	90	100	95	85	84	98	100

9. Calculate trend values by 4 yearly moving averages for the following data.

Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Production	464	515	518	467	502	540	557	571	586	612

10. Fit a straight line trend for the following data by least squares method and estimate the production for 2010.

Year	2003	2004	2005	2006	2007
Production (in lakhs)	70	74	81	86	89

11. Fit a straight line trend for the following data by least squares method and estimate the price for 2013.

Year	2007	2008	2009	2010	2011
Price (in Rs.)	12	20	31	40	47

12. For the following time series fit a linear trend by the method of least squares.

Year	1980	1984	1988	1992	1996	2000
Sales ('000)	16	20	19	14	18	21

13. Below are given the figures of production of a sugar factory. Fit a linear trend by the method of least squares.

Year	2010	2011	2012	2013	2014	2015	2016
Production ('000 tons)	79	83	90	94	90	95	99

Ten mark questions:

1. Production in 1000 quintals of sugar is given below. Fit a straight line trend, find trend values and estimate production for 2009.

Year	2002	2003	2004	2005	2006	2007	2008
Production	12	10	14	11	13	15	16

2. Fit a straight line trend for the following data by the method of least squares, find trend values and estimate the production for the year 2007.

Year	2000	2001	2002	2003	2004	2005	2006
Production ('000 tons)	50	47	52	45	48	55	60

3. For the following data fit a straight line trend of the form Y = a + b X. Find the trend values and estimate the sale for 2007.

Year	2000	2001	2002	2003	2004	2005	2006
Sale of Vehicles ('000)	160	350	340	580	770	910	950

4. Below are given the figures of production (in thousand tons) of a sugar factory

Year	2001	2002	2003	2004	2005	2006	2007
Production	80	90	92	83	94	99	92

(i) Fit a st. line by the method of least squares and obtain the trend values.
(ii) Also estimate the production for the year 2008.

5. Fit a straight line trend for the following data by the method of least squares, find trend values and estimate the production for 2015.

Year	2007	2008	2009	2010	2011	2012	2013
Production ('000 tons)	80	100	110	122	138	150	162

6. For the following time series fit a second degree trend of the type $y = a + bx + cx^2$ by the method of least squares. Estimate the profit for the year 2010.

Year	2005	2006	2007	2008	2009
Profit (in lakhs)	10	12	16	24	38

7. By the method of least squares fit a parabolic trend for the following time series. Estimate the profit for the year 2016.

Year	2011	2012	2013	2014	2015
Profit ('000)	10	12	14	10	8

8. By the method of least squares fit a parabolic trend for the following time series. Estimate the production for the year 2017.

Year	2011	2012	2013	2014	2015
production ('000 tons)	15	11	10	11	13

9. For the following time series data fit a trend equation of the type $y = a + bx + cx^2$. Estimate the value for 2011.

Year	2004	2005	2006	2007	2008	2009	2010
Value	14	16	20	29	42	60	80

10. The following figures give the annual production of a commodity. Estimate the production in 2011 by using the trend equation of the type $y = a + bx + cx^2$.

Year	2004	2005	2006	2007	2008	2009	2010
Production ('000 tons)	8	10	13	17	22	28	35

11. Population figures for a city are given below.

Year	2008	2009	2010	2011	2012
Population ('000)	132	142	157	170	191

Fit a curve of the type $y = ab^x$ and estimate the population for the year 2014

12. Following data shows the population of India. Fit a curve of the type $y = ab^x$ and estimate the population for the year 2011.

Year	1961	1971	1981	1991	2001
Population (in crores)	43.9	54.8	68.4	84.4	102.7

13. Following data shows the population of India. Fit a curve of the type $y = ab^x$. Estimate the population for the year 2021.

Year	1951	1961	1971	1981	1991	2001	2011
Population (in crores)	36.1	43.9	54.8	68.4	84.4	102.7	121

14. Following data shows the population of Karnataka. Fit a curve of the type $y=ab^x$. Estimate the population for the year 2021.

Year	1951	1961	1971	1981	1991	2001	2011
Population (in lakhs)	194	236	293	371	448	527	611

15. The sales of a company in lakhs of rupees for the year 2006 to 2012 are given below.

Year	2006	2007	2008	2009	2010	2011	2012
Sales	32	47	65	92	132	190	275

Fit an equation of the form $y = ab^x$ and estimate the sales for the year 2013.

16. For the following data fit an exponential trend of the form $y = ab^x$.

Year	2005	2006	2007	2008	2009	2010
Production (in crores)	7	10	12	14	17	24

UNIT – IV INTERPOLATION AND EXTRAPOLATION

Two mark questions:

1. Differentiate between interpolation and extrapolation.
2. Write two assumptions of interpolation and extrapolation.
3. Mention two methods of interpolation.
4. Write down the conditions for applying of Binomial expansion method of interpolation and extrapolation.
5. Write down the conditions for applying Newton's advancing difference method of interpolation.

Five mark questions:

1. Following is data regarding annual net life insurance premium. Using binomial expansion method estimate the premium at the age 30 and 45.

Age (in Years)	20	25	30	35	40	45
Premium (in Rs.)	1426	1581	-	1996	2256	-

2. For the following data estimate the production for the year 2001 and 2005 by binomial expansion method of interpolation and extrapolation.

Year	1995	1997	1999	2001	2003	2005
Production ('000 tons)	20	40	70	?	130	?

3. For the following data estimate the production for 2006 and 2010 by binomial expansion method of interpolation and extrapolation.

Year	2005	2006	2007	2008	2009	2010
Production (tons)	5	?	10	15	20	?

4. For the following data interpolate and extrapolate the sales for the year 2007 and 2011.

Year	2006	2007	2008	2009	2010	2011
Sales	13	?	25	38	65	?

5. Interpolate and extrapolate the production for the years 1989 and 1991 with the help of the following table.

Year	1986	1987	1988	1989	1990	1991
Production	120	122	126	-	135	-

6. The annual sales of a company are given below. Interpolate and extrapolate the sales for the year 2002 and for the year 2005.

Year	2000	2001	2002	2003	2004	2005
Sales (Rs.lakhs)	125	163	-	238	282	-

7. Interpolate and extrapolate the production for the years 1965 and 1975 with the help of the following table.

Year	1950	1955	1960	1965	1970	1975
Production (tons)	100	120	150	-	210	-

8. Interpolate and extrapolate the business for the years 1982 and 1984 from the following data.

Year	1978	1979	1980	1981	1982	1983	1984
Business (in lakhs)	80	150	235	365	-	780	-

9. Interpolate and extrapolate the production for the years 2000 and 2010 with the help of the following table.

Year	1980	1985	1990	1995	2000	2005	2010
Production (tons)	5	12	19	26	-	40	-

10. Cost of living indices of working class of a certain place for some years are given below.Interpolate and extrapolate the missing index number for 1996 and 1999.

Year	1993	1994	1995	1996	1997	1998	1999
Index No.	200	202	206	?	222	240	?

11. Interpolate and extrapolate the production for the years 1982 and 1985 from the following data.

Year	1979	1980	1981	1982	1983	1984	1985
Production (tons)	100	120	150	-	525	780	-

12. By binomial expansion method estimate the number of persons at ages 24 and 30 years with the help of the following data.

Age (Years)	18	20	22	24	26	28	30
No. of persons	20	22	26	?	35	39	?

13. Interpolate and extrapolate the missing values in the following the data.

X	5	10	15	20	25	30	35
Y	5	8	?	13	15	15	?

14. Following is data regarding annual net life insurance premium. Using Newton's advancing difference method estimate the premium at the age of 26 years.

Age (in Years)	20	25	30	35
Premium (in Rs.)	1426	1581	1771	1996

15. The following table shows the expectation of life at different ages. Interpolate the expectationof life at the age 15.

Age (in Years)	10	20	30	40
Expectation of life (in years)	50	42	33	22

16. Using Newton's forward difference method interpolate the value of 'y' when x=25.

X	10	20	30	40	50
Y	55	48	39	26	7

17. Below are given the wages earned by workers per day in a certain factory. Using Newton's advancing difference method estimate the number of workers earning up to Rs.550 per day.

Wages per day up to (Rs.)	500	600	700	800
No. of workers	50	150	300	500

18. Below are given the wages earned by workers per day in a certain factory. Using Newton's advancing difference method estimate the number of workers earning up to Rs.650 per day.

Wages per day up to (Rs.)	500	600	700	800	900
No. of workers	20	120	240	430	740

19. From the following data estimate the number of persons earning wages below Rs.70 per day using Newton's advancing difference method.

Wages per day	Below 40	40 - 60	60 - 80	80 - 100	100 - 120
No. of persons	500	280	200	140	100

20. From the following data interpolate the number of persons below the age 30 years.

Age (in Years)	Below 20	20 - 40	40 - 60	60 – 80	80 – 100
No. of persons ('000)	30	35	28	19	8

21. From the following table interpolate the number of workers getting salary below Rs.25000.

Wages ('000 Rs.)	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
No. of workers	50	80	100	75	40

UNIT – V THEORETICAL DISTRIBUTIONS

Choose the most appropriate answer from the choices given.

1. The range of Bernoulli distribution is
   a) x = 0,1 b) x = 0,1,2,3,.....n c) x = 0,1,2,3,....., ∞ d) x = ∞

2. The range of binomial distribution is
   a) x = 0,1 b) x = 0,1,2,3,.....n c) x = 0,1,2,3,....., ∞ d) x = ∞

3. The mean of a Bernoulli distribution is –
   a) p b) np c) pq d) npq

4. The variance of a Bernoulli distribution with parameter p is –
   a) p b) np c) pq d) npq

5. The standard deviation of a Bernoulli distribution is –
   a) p b) pq c) $\sqrt{pq}$ d) npq

6. The mean of a Binomial distribution is –
   a) p b) np c) pq d) npq

7. The variance of a Binomial distribution is –
   a) p b) np c) npq d) $\sqrt{npq}$

8. The standard deviation of a Binomial distribution is –
   a) p b) np c) npq d) $\sqrt{npq}$

9. Following is the probability distribution of a Binomial variate:

X	0	1	2	3	4
p(x)	0.0625	0.25	0.375	0.25	0.0625

   The mode of the distribution is
   a) 4 b) 2 c) 3 d) 1

10. In a Binomial distribution, if np =3, the relation between mean and mode is
    a) Mean = mode b) Mean> mode c) Mean < mode d) mean ≠ mode

11. In a binomial distribution, if n = 5 and p = 0.4, then the relation between mean and mode is
    a) Mean = mode b) Mean > mode c) Mean < mode d) mean ≠ mode

12. The range of Poisson distribution is
    a) x = 0,1 b) x = 0,1,2,3,.....n c) x = 0,1,2,3,....., ∞ d) x = ∞

13. The mean of a Poisson distribution is –
    a) p b) np c) λ d) $\sqrt{\lambda}$

14. The variance of a Poisson distribution is –
    a) p b) np c) λ d) $\sqrt{\lambda}$

15. The standard deviation of a Poisson distribution is –
    a) p b) np c) λ d) $\sqrt{\lambda}$

16. The relationship between mean and variance of a Poisson distribution is –
    a) Mean = variance b) Mean > variance
    c) Mean < variance d) Mean ≠ variance

17. In a Poisson distribution, if $\lambda = 1.2$, the value of the mode is
    a) 1 b) 0 c) 2 d) 1.2

18. In a Poisson distribution, if $\lambda = 2$, the values of the modes are
    a) 1, 2 b) 3, 4 c) 0, 4 d) 0, 3

19. In a Hyper geometric distribution, if a = 2, b = 4 and n = 3, the range is
    a) x = 0,1 b) x = 0,1,2 c) x = 0,1,2,3 d) x = 0,1,2,3,4

20. The area under the normal curve is –
    a) 1 b) 0.5 c) 0 d) -1

21. The relationship between variance and standard deviation of a standard normal distribution is –
    a) Variance = S.D. b) Variance > S.D
    c) Variance < S.D. d) Variance ≠ S.D.

22. If 'X' is a normal variate with mean 'μ' and S.D. 'σ', the value of $P(X < \mu)$ is –
    a) 0 b) 0.25 c) 0.5 d) 1

23. If 'X' is a normal variate with mean 'μ' and S.D. 'σ', the value of $P(X > \mu)$ is –
    a) 0 b) 0.25 c) 0.5 d) 1

24. If 'Z' is a standard normal variate , the value of $P(Z<0)$ is –
    a) 0 b) 0.25 c) 0.5 d) 1

25. If 'Z' is a standard normal variate , the value of $P(Z > 0)$ is –
    a) 0 b) 0.25 c) 0.5 d) 1

26. The parameter of a Chi- square distribution is –
    a) n b) p c) q d) n, p

27. The mean of a Chi-square distribution is –
    a) n b) p c) 2n d) np

28. The variance of a Chi-square distribution is –
    a) n b) p c) 2n d) $\sqrt{2n}$

29. The standard deviation of a Chi-square distribution is –
    a) n b) p c) 2n d) $\sqrt{2n}$

30. If Z_1, Z_2 and Z_3 are three independent standard normal variates, the distribution of $Z_1^2 + Z_2^2 + Z_3^2$ is –
    a) $\chi^2(1)$ b) $\chi^2(2)$ c) $\chi^2(3)$ d) $\chi^2(4)$

31. For a Chi-square distribution with 10 d.f., the mean is –
    a) 0 b) 5 c) 10 d) 20

32. The parameter of a t-distribution is-
    a) n b) p c) q d) n, p

33. The mean of a t- distribution is –
    a) 0 b) 0.5 c) 1 d) 2

34. The median of a t- distribution is –
    a) 0 b) 0.5 c) 1 d) 2

35. The mode of a t- distribution is –
    a) 0 b) 0.5 c) 1 d) 2

36. If parameter of a t-distribution is 10, the mean is –
    a) 0 b) 5 c) 10 d) 20

37. The variance of a t- distribution is-
    a) $\frac{n}{2-n}$ b) $\frac{n}{n+2}$ c) $\frac{n}{n-2}$ d) $\frac{n-2}{n}$

Fill in the blanks by choosing the appropriate word from those given in the brackets.

[0, 1, n, p, np, pq, npq, λ, 2n, Poisson, $\frac{1}{2}$, 0.5, Standard normal, t-distribution, Bell, 4, $\chi^2(3)$]

1. The mean of a Bernoulli distribution is _____.
2. The variance of a Bernoulli distribution is _____.
3. The mean of a Binomial distribution is _____.
4. The variance of a Binomial distribution is _____.
5. Binomial distribution is positively skewed when p <____.
6. Binomial distribution is negatively skewed when p >____.
7. The mean of a Poisson distribution is _____.
8. In _____distribution, mean and variance are equal.
9. The area under the normal curve is_____.
10. In _____distribution, variance and standard deviation are equal.
11. If Z_1, Z_2 and Z_3 are three independent standard normal variates, the distribution of ($Z_1^2 + Z_2^2 + Z_3^2$) is ____.
12. The mean of a Chi-square distribution is_____.
13. The variance of a Chi-square distribution is _____.
14. The mean of a t- distribution is _____.
15. If parameter of a t- distribution is 4, the mean is _____.
16. In a_____, mean = median = mode = 0
17. The t – curve is _____shaped.

Match the following: (Direct answers are given)

	Bernoulli distribution		Mean > variance
	Bernoulli distribution		Range: 0,1
	The range of Bernoulli distribution		x = 0,1
	Binomial distribution		Mean > variance
	Variance of a Bernoulli distribution		pq
	In a Poisson distribution, if $\lambda = 4$		Mean =4
	In a Poisson distribution, if $\lambda = 4$		Variance =4
	Poisson distribution		Mean = variance
	The area under the normal curve		1
	Standard normal distribution		Variance = standard deviation
	The parameter of a Chi-square distribution		n
	The mean of a Chi-square distribution		n
	The variance of a Chi-square distribution		2n
	Chi- square distribution		Mean equals to its parameter
	Chi-square distribution with 10 d.f.		Mean = 10
	Chi-square distribution curve		Leptokurtic ($\beta_2 > 3$)
	The parameter of a t- distribution		n
	The mean of a t- distribution		0
	The median of a t- distribution		0
	The mode of a t - distribution		0
	Student's t-distribution curve		Leptokurtic ($\beta_2 > 3$)

One mark questions:

1. Define a Bernoulli variate.
2. Define a Bernoulli distribution.
3. Write down the probability mass function of a Bernoulli distribution.
4. Write the range of a Bernoulli distribution.
5. Give an example for Bernoulli variate.
6. Write a feature of Bernoulli distribution.
7. Write the parameter of a Bernoulli distribution.
8. Write the mean of a Bernoulli distribution.
9. Write the variance of a Bernoulli distribution.
10. Write the relationship between mean and variance of a Bernoulli distribution.
11. If $X_1, X_2, \ldots X_n$ are independently identically distributed Bernoulli variates with parameter p, what is the distribution of $X=X_1+X_2+\ldots+X_n$?
12. If $p = \frac{4}{5}$ for a Bernoulli distribution, write down the p.m.f.
13. If $p = \frac{4}{5}$ for a Bernoulli distribution, find variance.
14. Define a binomial vaiate.
15. Define a binomial distribution.
16. Write down the p.m.f. of a binomial distribution.
17. Write down the range of a binomial distribution.
18. Give an example of binomial variate.
19. Write a feature of binomial distribution.
20. What are the parameters of binomial distribution?
21. Write the mean of a binomial distribution.

22. Write the variance of a binomial distribution.
23. Write the relationship between mean and variance of a binomial distribution.
24. Write the recurrence relation for successive probabilities in a binomial distribution.
25. Write the recurrence relation for successive frequencies in a binomial distribution.
26. For what value of 'p' binomial distribution is symmetrical?
27. From the following probability distribution find the value of mode.

x	0	1	2	Total
(x)	0.25	0.5	0.25	1

28. Define a Poisson variate.
29. Define a Poisson distribution.
30. Write down the p.m.f. of a Poisson distribution.
31. Write the range of a Poisson distribution.
32. Give an example for Poisson variate.
33. Mention a feature of Poisson distribution.
34. What is the parameter of a Poisson distribution?
35. Write the mean of a Poisson distribution.
36. Write the variance of a Poisson distribution.
37. Write the relationship between mean and variance of a Poisson distribution.
38. In which distribution mean and variance are equal?
39. Which type of skewness a Poisson distribution has?
40. Under what condition Poisson distribution tends to normal distribution?
41. In a Poisson distribution if mean is 4, find S.D.
42. In a Poisson distribution if S.D. is 4, find mean.
43. Write the recurrence formula for successive probabilities in a Poisson distribution.
44. Write the recurrence formula for successive frequencies in a Poisson distribution.
45. In a Poisson distribution, if $P(X = 0) = 0.0408$, find λ.
46. Define a hyper geometric variate.
47. Define a hyper geometric distribution.
48. Write down the p.m.f. of a hyper geometric distribution.
49. Write the range of a hyper geometric distribution.
50. Mention a feature of hyper geometric distribution.
51. Write the mean of a hyper geometric distribution.
52. Write the variance of a hyper geometric distribution.
53. Write down the p.m.f. of a hyper geometric distribution whose parameters are a=4, b=7 and n=5.
54. For a hyper geometric distribution the parameters are a = 7, b = 5 and n = 8. Write the range of hyper geometric variate.
55. Given, a = 3, b = 8 and n = 6, find the mean of a hyper geometric distribution.
56. Write down the mean of a hyper geometric distribution whose parameters are a=4, b=7 and n=5.
57. Define a normal variate.
58. Define a normal distribution.
59. Write down the p.d.f. of a normal distribution.
60. Write the range of a normal distribution.
61. Give an example for normal variate.
62. Write the mean of a normal distribution.
63. Write the variance of a normal distribution.
64. What is the value of coefficient of skewness (β_1) for a normal distribution?

65. What is the value of coefficient of kurtosis (β_2) for a normal distribution?
66. What is the probability that a normal variate takes a value greater than its mean?
67. What is the total area under the normal curve?
68. If X is a normal variate with mean μ and S.D.σ, what is the value of $P(X < \mu)$?
69. Write down the area under the normal curve in 1σ neighbourhood of μ.
70. Write down the area under the normal curve in 2σ neighbourhood of μ.
71. Write down the area under the normal curve in 3σ neighbourhood of μ.
72. Define a standard normal variate.
73. Define a standard normal distribution.
74. Write down the p.d.f. of a standard normal distribution.
75. Write the mean of a standard normal distribution.
76. Write the variance of a standard normal distribution.
77. Name the distribution for which variance and S.D. are equal.
78. If Z is a SNV, write the value of $P(Z < 0)$.
79. In a normal distribution, given $P(-0.8 < Z < 0.8) = 0.5762$. Find $P(0 < Z < 0.8)$.
80. If Z is a SNV and $P(Z > k) = 0.05$, find the value of k.
81. If Z is a SNV and $P(Z > k) = 0.1$, find the value of k.
82. Define a chi-square distribution.
83. Write down the p.d.f. of a chi-square distribution.
84. Write the range of a chi-square distribution.
85. Define degrees of freedom.
86. Mention a feature of chi-square distribution.
87. Write the parameter of a chi-square distribution.
88. Write the mean of a chi-square distribution.
89. Write the variance of a chi-square distribution.
90. If Z is a SNV, then name the distribution of Z2.
91. If Z_1, Z_2 and Z_3 are three independent standard normal variates, what is the distribution of $Z_1^2+Z_2^2+Z_3^2$?
92. Find mean of a chi-square variate with 8 degrees of freedom.
93. If variance of chi-square variate is 16, what is its mean?
94. Find mode of a chi-square variate with 8 degrees of freedom.
95. For a chi square(χ^2) variate, if $P(0<\chi^2<13.33) = 0.5$, find median.
96. Write down the p.d.f. of a student's t-distribution.
97. Write the range of a student's t-distribution.
98. Write the parameter of a student's t-distribution.
99. Mention a feature of student's t-distribution.
100. Write the mean of a t-distribution.
101. Write the variance of a t-distribution.

Two mark questions:

1. Write the mean and variance of a Bernoulli distribution.
2. What is meant by Bernoulli trial? Give an example.
3. Write down the Bernoulli distribution with parameter p = 0.25
4. Write down the Bernoulli distribution with parameter p =2/5.
5. Find the mean and variance of a Bernoulli distribution with parameter p = 0.8.
6. If p = 0.1 for a Bernoulli distribution, find mean and variance.
7. If $p = \frac{1}{2}$ for a Bernoulli distribution, find mean and variance.
8. Find the mean and variance of a Bernoulli distribution with parameter p = 2/3.

9. If $p = \frac{1}{5}$ for a Bernoulli distribution, write down the p.m.f.. Also find variance.
10. Write down the p.m.f. of a Bernoulli distribution with parameter 0.8. Also find S.D.
11. For a Bernoulli distribution, if p = 0.1 write the p.m.f. Also find S.D.
12. If n = 5 and p = 1/4, then write p.m.f. binomial distribution.
13. Write down two features of binomial distribution.
14. Write mean and variance a binomial distribution.
15. 'The mean and variance of a binomial distribution are 4 and 5 respectively'. Comment on this statement with reason.
16. Under which conditions binomial distribution tends to Poisson distribution?
17. Under which conditions binomial distribution tends to normal distribution?
18. In a binomial distribution, if n = 6 and p = 1/3, find mean and variance.
19. For a binomial distribution if n = 5 and p = 0.4, find mean and variance.
20. In a binomial distribution, if n = 6 and p = 1/3, find mean and mode.
21. For a binomial distribution if n = 5 and p = 0.4, find mean and mode.
22. In a binomial distribution, if n = 6 and p = 1/3, find mode and variance.
23. For a binomial distribution if n = 5 and p = 0.4, find mode and variance.
24. In a binomial distribution with 9 trials, the mean is 6. Find 'p' and the variance.
25. In a binomial distribution, if n = 16 and mean = 4. Find p and S.D.
26. If a binomial distribution has mean 3 and variance 2, find the parameters.
27. The mean and S.D. of a binomial distribution are 8 and 2 respectively. Find the parameters.
28. For a binomial distribution, if mean = 4 and S.D. = $\sqrt{2}$, find the parameters.
29. Mention two features of Poisson distribution.
30. Find P(X = 0) in a Poisson distribution with mean 5.
31. The probability that a taxi driver makes no accidents is 0.1353. Find the average number of accidents made by him.
32. In a Poisson distribution if P(X = 0) = 0.1225, write down the p.m.f.
33. In a Poisson distribution if 3×p(3) = p(4). Find mean.
34. In a Poisson distribution, the second probability term is half of the first probability term. Find the variance.
35. In a Poisson distribution the first probability term is 0.3679. Find the next probability term.
36. Mention two features of hyper geometric distribution.
37. Write the parameters of a hyper geometric distribution.
38. Write the mean and variance of a hyper geometric distribution.
39. Given, a = 5, b = 3 and n = 3. Find P(X = 1) for hyper geometric distribution.
40. In a hyper geometric distribution if a = 6, b = 9 and n = 4, find P(X = 2).
41. Find the variance of a hyper geometric distribution with parameters a = 12, b = 8 and n = 6.
42. If a = 5, b = 15 and n = 3, then find the variance of hyper geometric distribution.
43. Given a = 6, b = 4 and n = 4, find the S.D of hyper geometric distribution.
44. For a hyper geometric distribution the parameters are a = 7, b = 5 and n = 8. What is the S.D?
45. Under what conditions hyper geometric distribution tends to binomial distribution?
46. Write the parameters of a normal distribution.
47. Write the mean and variance of a normal distribution.
48. Write the p.d.f. of a normal distribution with mean 3 and S.D. 2
49. Write the p.d.f. of a normal distribution with mean 10 and S.D. 3
50. Write the p.d.f. of a normal distribution with mean 55 and variance 4.

51. Consider the following p.d.f. of a normal variate X, $f(X) = \frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-6}{3}\right)^2}$; $-\infty < x < \infty$. Write the mean and variance of X.
52. The distribution of a variable X is given by, $f(X) = \frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-100}{5}\right)^2}$; $-\infty < x < \infty$. Write the mean and S.D.
53. What are the values of coefficient of skewness (β_1) and coefficient of kurtosis (β_2) for a normal distribution?
54. The two parameters of a normal distribution are μ= 10 and σ = 5. Find the points of inflexion of the curve.
55. Mean and variance of a normal distribution are 12 and 4 respectively. Find points of inflexion.
56. If the variance of a normal distribution is 9 cm^2., then find Q.D.
57. If the S.D. of a normal distribution is 15, find M.D.
58. In a normal distribution if the values of first and third quartiles are 25 and 55 respectively, find the mean.
59. If the lower and upper quartiles of a normal distribution are 11 and 39 respectively, find the median.
60. If Q_1 = 30 and Q_3 = 70, find the mode of the normal distribution.
61. In a normal distribution mean and variance are 50 and 16 respectively. Find the Q_1 and Q_3.
62. Mean and S.D. of a normal distribution are 20 and 2 respectively. Find lower and upper quartiles.
63. In a normal distribution Q_1 and mean are 40 and 50 respectively. Find the S.D.
64. In a normal distribution mean and Q_3 are 50 and 60 respectively. Find the S.D.
65. In a normal distribution Q_1 and S.D. are 40 and 15 respectively. Find the mean.
66. In a normal distribution Q_3 and S.D. are 60 and 15 respectively. Find the mean.
67. Write the mean and variance of a standard normal distribution.
68. If Z is a SNV and P(Z < k) = 0.95, find the value of k.
69. If Z is a SNV, then name the distribution of Z^2 and write its variance.
70. If Z_1 and Z_2 are two independent SNVs, then name the distribution of $Z_1^2+Z_2^2$ and write its mean.
71. Mention two features of chi-square distribution.
72. Write the mean and variance of a chi-square distribution.
73. What are the mean and variance of chi-square variate with 18 d.f.?
74. Find the S.D. of a chi-square variate with 8 d.f.
75. If variance of a chi-square variate is 16, what are mean and mode?
76. For a chi square(χ^2)variate, with 13 d.f. $P(0<\chi^2<12.33)$ = 0.5. Find median and mode.
77. For a chi square(χ^2)variate, with 10 d.f.$P(0<\chi^2<9.33)$ = 0.5. Find median and mode.
78. Mention two features of student's t-distribution.
79. Write the mean and variance of a student's t-distribution.
80. If the parameter of t-distribution is 6, find mean and variance.
81. If n = 4 for student's t-distribution, find S.D.
82. If the parameter of t-distribution is 7, find median and variance.
83. If n = 10 for student's t-distribution, find mode and variance.

Five mark questions:

1. Assuming that birth to male child and birth to female child to be equi-probable, find the probability that a family with three children has 2 or more male children.

2. An unbiased coin is tossed 5 times. Find the probability that the tosses results in i) heads only ii) 3 heads.

3. An unbiased coin is tossed 5 times. Find the probability that the tosses results in i) one head ii) 4 or more heads.

4. The probability that a bomb hits a target is 1/2. Four bombs are aimed at a bridge.3 bombs are enough to destroy the bridge. What is the probability that i) the bridge is destroyed ii) none of the bombs hit the bridge.

5. Team A has probability 0.3 of winning a game. If it plays 5 games, find the probability that it wins i) 4 games ii) at least one game.

6. In a college, 60% of the students are boys. In a random sample of 3 students, calculate the probability of getting i) two boys ii) at least one boy.

7. There are 40 boys and 30 girls in a class. 4 students are selected at random. Find the probability that the selected students are i) all boys ii) at the most two boys.

8. In a college 35% of the students are girls. In a random sample of 5 students, calculate the probability that i) two are girls, ii) at least one is girl.

9. If the chance that a vessel (ship) arrives safely at a port is 9/10. Find the chance that out of 5 vessels, at least 4 will arrive safely.

10. The probability of an arrow hitting a tree is 1/3. If 4 arrows are aimed at the tree, find the probability that i) 3 arrows miss the tree ii) at least two arrows hit the tree.

11. In a certain university, the chance that a professor suffers from dust allergy is 1/5. Calculate the chance that out of 8 professors in a university i) 2 will not suffer from the allergy, ii) at least 2 will suffer from the allergy.

12. In a city 40% of the people are vegetarians. In a random sample of 6 persons find the probability that i) 3 are vegetarians, ii) at least one is vegetarian.

13. The incidence of an occupational disease in an industry is such that the workers have 20% chance of suffering from it. What is the probability that out of 5 workers 4 or more suffer from the disease?

14. The incidence of an occupational disease in an industry is such that the workers have 20% chance of suffering from it. What is the probability that out of 5 workers at most 1 suffer from the disease?

15. There are 100 wrist watches in a box. 5 of them are defective. In a selection of 6 watches, what is the probability of getting less than 2 defective wrist watches? If there are 50 such boxes, in how many of them will you find 2 defective wrist watches?

16. In a grove there are 200 trees. Out of which 50 are mango trees. Among them, if 20 samples of 3 trees each are selected, in how many samples will you expect i) exactly one mango tree ii) more than one mango tree?

17. In a village 1/3 of the people are literates. If 100 investigators meet 5 persons each to see if they are literates, then how many investigators would you expect to report that 2 or more are literates?

18. In a certain school 40% of the students have opted for first language Kannada. Assuming 20 teachers take a sample of 4 students each, how many teachers will report that 2 or 3 students opted for first language Kannada?

19. Four unbiased coins are tossed 64 times. Find expected frequencies of number of heads obtained.

20. Four unbiased coins are tossed 128 times. Find the theoretical frequencies for the number of heads obtained.

21. Five unbiased coins are tossed 128 times. Calculate the expected frequencies for the number of heads obtained.

22. Five unbiased coins are tossed 256 times. Find the theoretical frequencies for the number of tails obtained.

23. Six unbiased coins are tossed 256 times. Calculate the expected frequencies for the number of heads obtained.

24. Fit a binomial distribution to the following data and obtain the theoretical frequencies.

Number of defective balloons	0	1	2	3	4
Number of packets	6	12	22	24	16

25. Fit a binomial distribution to the following data and obtain the expected frequencies.

Male births	0	1	2	3	4
No. of families	46	194	270	230	60

26. Four coins are tossed 100 times and the following distribution is obtained.

Number of heads	0	1	2	3	4
Number of tosses	8	32	33	22	5

Fit a binomial distribution when nature of the coin is not known. Obtain the theoretical frequencies.

27. Five coins are tossed 256 times and the following distribution is obtained.

Number of heads	0	1	2	3	4	5
Number of tosses	35	39	45	62	50	25

Fit a binomial distribution when nature of the coin is not known. Obtain the theoretical frequencies.

28. For the following data, fit a binomial distribution and obtain the expected frequencies.

Number of defective screws	0	1	2	3	4
Number of samples	3	20	26	17	9

29. The following data relates to the number of defective items.

Number of defective items	0	1	2	3	4	5
Number of samples	171	200	99	20	8	2

Fit a binomial distribution to the data. Obtain the theoretical frequencies.

30. In a binomial distribution with parameters n = 3 and p, the second and the third frequency terms are 26 and 24 respectively. Find p

31. A typist makes 2mistakes per page on an average. Find the probability that a page typed by him has i) 2 mistakes ii) at the most 2 mistakes.

32. On an average, a telephone operator receives 3 telephone calls per 5 minutes interval. Calculate the probability that in a particular 5 minutes interval he i) does not receive any call ii) receives more than one call.

33. On an average a box contains 2 defective items. Find the probability that a randomly selected box has i) no defective item ii) at least two defective items.

34. It has been found that on an average 4 patients visit a particular doctor during one hour. What is the probability that during a particular hour i) doctor is free ii) more than 3 patients visit the doctor?

35. A car hire agency has two cars. On an average there is a demand for one car during a particular hour. What is the probability that i) both the cars are free ii) some demand is refused?

36. If 2% of electric bulbs manufactured by a company are known to be defective, what is the probability that a sample of 150 electric bulbs taken from the production process of that company would contain i) exactly one defective bulb? ii) more than two defective bulbs? (Use Poisson approximation).

37. A 200 pages book has 0.4 mistakes per page on an average. i) Find the probability that a randomly selected page from the book is free from mistakes ii) in the book how many pages contain mistakes?

38. In a text book, on an average 0.3 mistakes per page is found. If there are 500 pages in the text book, in how many pages will there be i) three mistakes ii) at the most two mistakes?

39. On an average, the number of defective items in a box is 4. If there are 100 such boxes, in how many of them would you expect i) one defective item ii) at least 2 defective items?

40. The number of accidents in a year attributed to taxi drivers in a city follows Poisson distribution with mean 2.5. Out of 1000 taxi drivers, find approximate number of drivers with i) one accident in a year ii) more than three accidents in a year.

41. On an average, one in every 50 valves manufactured by a firm is substandard. If valves are supplied in packets of 100 each, in how many of the lot of 2000 packets would you expect substandard valves? (Use Poisson approximation).

42. The probability that a knife manufactured by a firm is defective is 1/50. Knives are supplied in boxes of 50 each. In a lot of 1000 boxes, how many boxes contain i) 3 defective knives ii) at least one defective knife? (Use Poisson approximation).

43. Out of experience, it is known that 1% of the screws manufactured by a firm are defective. Screws are supplied in packets of 100 each. What is the probability that a randomly selected packet has 2 defective screws? Among 3000 packets, in how many packets would you expect defective screws? (Use Poisson approximation).

44. The probability of a fuse manufactured by a firm found to be defective is 0.05. Find the probability that a box containing 100 fuses have no defective fuse. Among 500 such boxes how many contain exactly 4 defective fuses? (Use Poisson approximation).

45. In a certain factory turning out optical lenses there is a chance 1/500 for lens to be defective. The lenses are supplied in packets of 50 each. Use Poisson distribution to calculate the approximate number of packets containing one defective lens in a consignment of 20000 packets.

46. If a random variable X follows Poisson distribution such that P(X=1) = P(X = 2), find i) mean ii) standard deviation and iii) the first probability term.

47. In a Poisson distribution, P(X = 2) = P(X = 3). Find P(X = 4).

48. In a Poisson distribution, if first two frequencies are 100 and 120 respectively. Find next two frequencies.

49. In a Poisson distribution, the second and third frequencies are 120 and 60 respectively. Find next two frequencies.

50. The following are mistakes per page observed in a book.

Number of mistakes per page	0	1	2	3	4 and more
Number of pages	68	37	10	5	0

Fit a Poisson distribution to the data. Obtain expected frequencies.

51. Fit a Poisson distribution to the following data and hence find the theoretical frequencies.

No. of deaths per day	0	1	2	3	4 and more
No. of days	20	45	30	5	0

52. Fit a Poisson distribution to the following data and hence find the theoretical frequencies.

No. of T.V. Sets sold	0	1	2	3	4	5 and more
No. of days	31	34	21	12	2	0

53. Fit a Poisson distribution to the following data and hence find the theoretical frequencies.

X	0	1	2	3	4	5
f	46	29	12	7	4	2

54. A bag contains 10 red and 5 black marbles. A random sample of 5 marbles is taken. Find the probability that the sample contains 3 red marbles. Also find the mean of red marbles.

55. There are 14 fruits in a basket, out of which 8 are mangoes and rests are oranges. A girl picks 5 fruits at random from the basket. Find the probability that she gets 3 mangoes. Also find the mean of mangoes.

56. A student preparing for an examination studies only 10 out of 12 sections prescribed. If the teacher selects 8 sections at random, what is the probability that the student will have studied i) six and ii) five of these sections?

57. There are 15 lecturers in a college. Out of them 8, belong to the science faculty. The college management builds 5 residential quarters and allots them to 5 randomly selected lecturers. Find the probability that i) all the quarters are allotted to science lecturers.ii) three quarters are allotted to science lecturers.

58. A pond has 10 fishes among which 4 are marked (marked fishes are under scientific observation). 4 fishes are caught from the pond. Find the probability that two of them are marked. Also find the mean of marked fishes.

59. A basket has 10 mangoes, out of which 6 are ripe. 3 mangoes are randomly selected.
    i) What is the probability that all the picked mangoes are ripe?
    ii) Find the expected number of ripe mangoes among the picked ones.

60. In a normal distribution, $Q_1 = 40$ and $Q_3 = 60$. Find mean, Q.D. and S.D.

61. In a normal distribution, mean and S.D. are 50 and 15 respectively. Find Q_1, Q_2 and Q_3.

62. If X is a normal variate with mean 64 and S.D. 4, find the probability that,
    i) $X \geq 68$ ii) $60 \leq X < 68$

63. If X is a normal variate with mean 64 and S.D. 4, find the probability that,
    i) $X < 62$ ii) $60 \le X < 66$
64. If X is a normal variate with mean 64 and variance 4, find the probability that,
    i) $X \ge 60$ ii) $X < 66$.
65. If $X \sim N(50, 3^2)$, find the probability that the value of the variate X is,
    i) more than 47, ii) lies between 44 and 53.
66. Marks scored by the students of a class follow normal distribution with mean 80 and S.D. 5. Calculate the probability that a student selected at random from the class scored i) more than 90 marks ii) between 70 and 85 marks.
67. Marks scored by the students of a class follow normal distribution with mean 80 and S.D. 5. Find the probability that a student selected at random from the class scored i) more than 70 marks ii) between 75 and 90 marks.
68. Monthly income of employees follows normal distribution with mean Rs.18,000 and S.D. Rs.800. Compute the probability of employees with monthly income i) more than Rs.20,000 ii) lies between Rs.16,000 and Rs.17,000.
69. The weekly wages of workers are normally distributed with mean Rs.3000 and S.D. Rs.500. Calculate the probability of workers whose weekly wages will be i) more than Rs.3400 ii) between Rs.2500 and Rs.3500.
70. The weekly wages of workers are normally distributed with mean Rs.3000 and S.D. Rs.500. Find the probability of workers whose weekly wages will be i) more than Rs.2500 ii) between Rs.2600 and Rs.3200.
71. The mean I.Q. of a large number of children of age fourteen is 95 and the S.D. is 5. Assuming that I.Q. follows normal distribution, find the i) percentage of the children with I.Q. under 85. ii) proportion of the children with I.Q. above 90.
72. The weights of 1000 students are normally distributed with mean 55 kg and S.D. 2 kg. Calculate the number of students with weight between 57 kg and 60 kg.
73. Heights of 2000 soldiers are normally distributed with mean 175 cm and variance 16 cm^2. Find the number of soldiers with height between 173 cm and 177 cm.
74. Heights of 360 children are normally distributed with mean 120 cm. and variance 4 cm^2. Find the number of children having height i) greater than 118 cm. ii) less than 124 cm.
75. The weights of 1000 youths are normally distributed with mean 60 kg and S.D. 4 kg. Find the number of youths with weight i) more than 56 kg. ii) less than 68 kg.
76. The weights of 500 persons are normally distributed with mean 60 kg and S.D. 5 kg. Find the number of persons with weight between 62 kg and 70 kg.
77. The weights of 500 persons are normally distributed with mean 60 kg and S.D. 5kg. Find the number of persons with weight i) more than 68 kg ii) less than 55 kg.
78. Height of a group of 500 candidates who attended the army selection camp is normally distributed with mean 170cm and S.D. 5cm. Minimum eligible height for army selection is 175cm.How many candidates from above group are eligible?
79. Height of a group of candidates who attended the army selection camp is normally distributed with mean 170cm and S.D. 3.9cm. Minimum eligible height for army selection is 175cm. Show that 10% of the above group is eligible.
80. The daily wages of workers are normally distributed with mean Rs.500 and S.D. Rs.50. i) Find the probability of workers whose daily wages will be between Rs.400 and Rs.600 ii) Show that only 10% of workers have daily wages more than Rs.564.

UNIT – VI STATISTICAL INFERENCE

Choose the most appropriate answer from the choices given.

1. A statistical constant of the population is called:
   a) A parameter b) A statistic c) Standard error d) Estimate

2. A function of the sample values is called:
   a) A parameter b) A statistic c) Standard error d) Estimate

3. The standard error of the sample mean is –
   a) $\sigma/\sqrt{n}$ b) $n/\sqrt{\sigma}$ c) $\sqrt{\sigma}/n$ d) σ/n

4. The standard error of sample proportion is-
   a) $\frac{PQ}{n}$ b) $\frac{P}{n}$ c) $\sqrt{\frac{PQ}{n}}$ d) $\frac{q}{n}$

5. The likely value of the unknown parameter is
   a) Estimate b) Estimator c) Estimation d) Point estimation

6. Statistic which is used to estimate an unknown parameter is called
   a) Estimate b) Point estimate c) Estimation d) Estimator

7. The best estimator of the population mean is-
   a) μ b) $\bar{x}$ c) σ d) s

8. Type I error is
   a) Rejecting H_0 when it is true b) Accepting H_0 when it is true
   c) Rejecting H_0 when it is not true d) Accepting H_0 when it is not true

9. The error that occurs by rejecting the null hypothesis when it is actually true is called:
   a) Size of the test b) Power of a test c) Type – I error d) Type – II error

10. The error that occurs by accepting the null hypothesis when it is actually not true is called:
    a) Size of the test b) Power of a test c) Type – I error d) Type – II error

11. The probability of rejecting the null hypothesis, when it is true is called:
    a) Size of the test b) Power of a test c) Type – I error d) Type – II error

12. The probability of rejecting the null hypothesis, when it is not true is called:
    a) Size of a test b) Power of a test c) Type – I error d) Type – II error

13. The degrees of freedom while testing independence of attributes in a 2x2 contingency table is –
    a) 1 b) 2 c)3 d) 4

14. A wrong decision about null hypothesis (H_0) leads to
    a) One kind of error b) Two kinds of error
    c) Three kinds of error d) Four kinds of error

15. There are four possible decisions under the testing of null hypothesis (H_0):
    i) Accept H_0 when it is true. ii) Reject H_0 when it is not true.
    iii) Accept H_0 when it is not true. iv) Reject H_0 when it is true.
    The correct decisions are:
    a) i and ii b) iii and iv c) i and iii d) ii and iii

16. There are four statements regarding the probability of rejecting null hypothesis (H_0):
    i) Size of the test = P(type – I error) ii) Size of the test = 1 – P(type – I error)
    iii) Power of a test = P(type – II error) iv) Power of a test = 1– P(type – II error)
    The correct statements are:
    a) i and ii b) iii and iv c) i and iv d) ii and iii

17. Which of the following statements are correct?
    i) P(type – I error) = α ii) P(type – II error) = β
    iii) P(type – I error) = 1 – α iv) P(type – II error) = 1– β
    a) i and ii b) iii and iv c) i and iv d) ii and iii

Fill in the blanks by choosing the appropriate word from those given in the brackets.
[Estimation, Interval estimation, Statistic, Test statistic, An estimator, An upper, Point estimation, Less than 5, Parameter space, Parameter, $\bar{x}$, Confidence limits, Size of the test, Type – I error, Power of a test, Critical region, Type – II error, $\sqrt{\frac{PQ}{n}}$, Acceptance region, Null distribution]

1. A statistical constant of the population is called a_____.
2. A function of the sample values is called a_____.
3. The set of all the admissible values of the parameters is called _____.
4. The standard deviation of the sampling distribution of a statistic is called its _____.
5. Standard error of sample proportion (p) is _____.
6. Method of obtaining the most likely value of the population parameter is called _____.
7. Any statistic which is used to estimate an unknown parameter is called_____.
8. If a single value is proposed as an estimate of the unknown parameter, it is _____.
9. If an interval is proposed as an estimate of the unknown parameter, it is _____.
10. The boundary values of confidence interval are _____.
11. The best estimator of the population mean is _____.

12. The error that occurs by rejecting the null hypothesis when it is actually true is called _____.
13. The error that occurs by accepting the null hypothesis when it is actually not true is called _____.
14. The probability of rejecting the null hypothesis, when it is true is called _____.
15. The probability of rejecting the null hypothesis, when it is not true is called _____.
16. _____is the set of those values of test statistic, which leads to the rejection of the null hypothesis.
17. _____is the set of those values of test statistic, which leads to the acceptance of the null hypothesis.
18. The statistical distribution of the test statistic under the null hypothesis is called_____.
19. A statistic based on whose distribution the testing of hypothesis is conducted is called_____.
20. Chi-square test for goodness of fit is always _____tailed test.
21. In chi-square test for goodness of fit, if any E_i is_____, it should be pooled with the adjacent frequency.

Match the following. (Direct answers are given)

	Parameter		Population constant
	Statistic		A function of sample values
	The best estimator of the population mean		$\bar{x}$ (sample mean)
	$\sigma/\sqrt{n}$		S. E.($\bar{x}$)
	Estimation and testing of hypothesis		Branches of statistical inference
	The likely value of the unknown parameter		Estimate
	Single value proposed as an estimate		Point estimation
	H_1: $\mu < 5$		H_0: $\mu = 5$
	H_1: $\mu > 5$		H_0: $\mu = 5$
	H_1: $\mu \neq 5$		H_0: $\mu = 5$
	P(type – I error)		α
	P(type – II error)		β
	Size of the test		P(Reject H_0 when it is true)
	Size of the test		P(type – I error)
	Power of a test		P(Reject H_0 when it is not true)
	Power of a test		1 – P(type – II error)
	Power of a test		$1 - \beta$

One mark questions:

1. Define parameter.
2. Define statistic.
3. What is parameter space?
4. What is sample space?
5. Define sampling distribution of a statistic.
6. Define standard error.
7. Write the formula of S.E.($\bar{x}$).
8. Write the formula of S.E.($\bar{x}_1 - \bar{x}_2$).
9. Write the formula of S.E. (p).
10. Write the formula of S.E.(p_1 - p_2), when $P_1 \neq P_2$.
11. Write the formula of S.E.(p_1 - p_2), when $P_1 = P_2$.
12. A lot contains 2% defective items. 40 items chosen from it. Another lot contains 1% defective items. 60 items chosen from it. Find E (p_1-p_2).
13. Write a use of standard error.
14. What is statistical inference?
15. Mention a branch of statistical inference.
16. What is meant by estimation?
17. Define an estimator.
18. Define an estimate.
19. Define point estimation.
20. Define interval estimation.
21. Define confidence interval.
22. What are confidence limits?
23. What is confidence coefficient?
24. Define statistical hypothesis.
25. Give an example for statistical hypothesis.
26. Define null hypothesis.
27. Give an example for null hypothesis.
28. Define alternative hypothesis.
29. Give an example for alternative hypothesis.
30. What is type I error?
31. What is type II error?
32. Define size of a test.
33. What is level of significance?
34. Define power of a test.
35. What is critical region?
36. What is critical value?
37. What is two tailed test?
38. What is one tailed test?
39. Define null distribution.
40. Define test statistic.

Two mark questions:

1. Define parameter and statistic.
2. Given $\sigma^2 = 9$ cm^2 and n = 36, calculate standard error of sample mean.
3. Sizes of two samples are 50 and 100. Population standard deviations are 20 and 10. ComputeS.E.($\bar{x}_1 - \bar{x}_2$).
4. If P = 0.02 and n = 64, then find S.E(p).
5. If P = 0.1 and n = 100, then find S.E(p).
6. A lot contains 2% defective items. 40 items chosen from it. Another lot contains 1% defective items. 60 items chosen from it. Find S.E.(p_1 - p_2).

7. For the following data, Find S.E.(p_1 - p_2).

Sample sizes	100	100
Population proportions	0.2	0.1

8. Write two uses of standard error.
9. Mention two branches of statistical inference.
10. Mention two types of estimation.
11. What are point and interval estimations?
12. What is statistical hypothesis? Give an example.
13. What is null hypothesis? Give an example.
14. What is alternative hypothesis? Give an example.
15. Define type I and type II errors.
16. Define size of a test and power of a test.
17. What are one tailed and two tailed tests?

Unit VI (b)
Large Sample Tests

One mark questions:

1. Write an alternative hypothesis for null hypothesis: $\mu = \mu_0$.

Two mark questions:

1. Given $\bar{x}$ = 203 gm, μ = 200 gm, σ = 10 gm and n = 64, calculate test statistic Z.
2. If $(\bar{x}_1 - \bar{x}_2)$= 2.6 and S.E.$(\bar{x}_1 - \bar{x}_2)$= 1.3, find test statistic Z.

Five mark questions:

1. A sample of 100 students is chosen from a large group of students. The average height of these students is 162 cm and standard deviation is 8 cm. At 5% level of significance can we reasonably assume that the average height of large group of students is 160 cm?

2. A random sample of 400 tins of Vanaspati has mean weight 4.96 kg and standard deviation 0.4 kg. Test at 1% level of significance that the average weight of tins of Vanaspati is 5 kg?

3. It is required to test whether those who practice Yoga have average blood sugar less than 120 mg/dl. A sample consisting of 36 persons who practice Yoga is observed. If their mean blood sugar is 118.5 mg/dl and variance is 9 mg^2/dl. At 1% level of significance what would you conclude?

4. The average weekly sales of the chocobars in candy stores were 150 bars. After advertising campaign the average weekly sales in 64 stores for a typical week increased to 153 bars and standard deviation 10 bars. Is the advertisement campaign successful? (Use α = 0.01)

5. A sample of 100 students is taken from a college. If the mean and S.D. of their weights are 51 kg and 5 kg respectively, test at 1% level of significance that the average weight of college students is 50 kg.

6. A machine is designed to fill 500 ml of milk to polythene bags. A randomly selected 100 milk bags filled by this machine are inspected. The mean is found to be 499 ml and S. D. is 5 ml. Is the machine functioning properly at 5% level of significance?

7. A random sample of 64 children is taken from a school. The average weight of the children is 29 kg and standard deviation is 5 kg. Can we assume that the average weight of the school children is less than 30 kg? (Use $\alpha = 0.05$)

8. A company manufactures car tyres. Their average life is 40,000 kilometres and standard deviation 5,000 kilometres. A change in the production process is believed to result in a better product. A test sample of 100 new tyres has mean life of 41,000 kilometres. Can you conclude at 5% L.O.S. that the new product gives better result?

9. A specified brand of automobile tyre is known to have average life of 10,000 km with a S.D. of 500 km. A random sample of 36 tyres, when tested resulted in the average life of 9,800 km. At 1% level of significance what is your conclusion regarding average life of tyres?

10. A random sample of 225 cans containing baby food has mean weight 998 gm and S. D. 15 gm. Test whether the mean contents of the cans be considered as 1 kg. Use 1% L.O.S.

11. On 60 different days the number of passengers in a bus was noted. The mean and S. D. of the number of passengers was found to be 40 and 5 respectively. At 5% level of significance, test the hypothesis that the average number of passengers in the bus is more than 38.

12. From the following data test whether mean daily wages of workers of Factory-A and Factory-B are equal. Use 1% L.O.S.

Factory	No. of workers	Mean daily wages (in Rs.)	S.D. (in Rs.)
A	200	195	20
B	450	200	30

13. The mean I.Q. of 200 randomly selected boys of a college is 90 and that of 128 girls is 88. Standard deviations of their I.Q. are 10 and 8 respectively. Test whether there is a significant difference between average I. Q. of boys and girls at 5% level of significance.

14. The mean and variance of heights of 60 randomly selected Punjabis are 178 cm and 9 cm^2 respectively. The mean and variance of heights of 40 randomly selected Guajaratis are 176 cm and 16 cm^2 respectively. Can we conclude that Punjabis are taller than Guajaratis? Use $\alpha = 5\%$.

15. For the following data, test whether means differ significantly. Use $\alpha = 0.01$.

Sample	Size	Mean	S.D.
I	90	52	9
II	40	54	2

16. 450 boys and 350 girls appeared for II P.U.C examination. The mean and S.D. of marks obtained by boys are 53 and 18 respectively. The mean and S.D of marks obtained by girls are 50 and 14 respectively. Is there any significant difference between mean marks obtained by boys and girls? (Use $\alpha = 0.01$)

17. A random sample of 100 workers from South India shows that their daily mean wage is Rs. 446 with S.D. Rs. 20. A random sample of 150 workers from North India shows that their daily mean wage is Rs. 450 with S.D. Rs. 30. Test at 1% level of significance that, mean wages in South India is less than mean wages in North India.

18. The mean and S.D. of marks obtained by 50 students of college A are 74 and 8 respectively. The mean and S.D. of marks obtained by 40 students of college B are 78 and 7 respectively. Is there any significant difference between mean marks of the students of two colleges? Test at 1% L.O.S.

19. The mean weight of 50 boys of college A is 58 kg with variance 64 kg^2. The mean weight of 40 boys of college B is 54 kg with variance 49 kg^2. Can we conclude that boys of college A and boys of college B have same mean weight? Use 5% level of significance.

20. A sample of 400 woman shoppers is chosen from market A. Their average weekly expenditure on food is Rs. 1000 with S.D. of Rs. 40. Another sample of 500 woman shoppers is chosen from market B. Their average weekly expenditure on food is Rs. 992 with S.D. of Rs. 50. Test at 5% level of significance that the average weekly expenditure on food is same.

21. Intelligence test given to groups of boys and girls gave the following information:

	Mean	S.D.	Size
Boys	70	12	100
Girls	74	10	50

Is the difference in the mean scores of boys and girls statistically significant? Use 5% L.O.S.

22. For the following data, test whether means differ significantly. Use $\alpha = 0.01$

Sample	Size	Mean	S.D.
I	40	70	8
II	60	66	6

23. Test at 1% level of significance, that average life of bulbs manufactured by Firm-A is less than Firm-B.

Firm	Size	Average life	Variance
A	32	1300 hours	64 $hours^2$
B	50	1305 hours	100 $hours^2$

24. Following is data regarding mean weights of randomly selected boys and girls of P.U.C. Test whether, mean weight of boys is greater than mean weight of girls. (Use $\alpha = 0.05$)

Sample	Boys	Girls
Size	64	48
Mean	63 kg	60 kg
S.D	8 kg	12 kg

25. Intelligence test given to groups of boys and girls gave the following information:

	Mean	S.D.	Size
Boys	74	12	100
Girls	70	10	50

Is the difference in mean score of boys and girls statistically significant? Use 1% L.O.S

26. In a sample of 400 people from Kerala 180 are coffee drinkers and the rest are tea drinkers. Can we assume that both coffee and tea are equally popular in the state? Use 5% level of significance.

27. A campus bulletin claims that only 36% of all college students visit the library in a week. A sample of 225 college students showed that 90 of them had visited the library. Can we accept the bulletin claim? (Use 1% L.O.S)

28. From a random sample of 100 students from PUC, 13 students were found wearing spectacles. Can we conclude that proportion of students wearing spectacles is more than 0.1? Use $\alpha = 0.05$.

29. A coin is tossed 400 times and head turns up 220 times. Can we conclude that the coin is unbiased? Use $\alpha = 0.05$.

30. In a city, out of 900 men 486 were smokers. Does this information indicate that the majority of men in the city are smokers? Use 1% L.O.S.

31. A manufacturer claims that less than 2% of his products are defective. A retailer buys a batch of 1000 articles from the manufacturer and finds that 10 are defective. Test at 5% level of significance that, whether the manufacturer's claim is justifiable.

32. A stock broker claims that he can predict with 75% accuracy whether a stock market value will rise or fall during the coming month. In a sample of 50 predictions he is correct in 35. Does this evidence supports broker's claim at 5% level of significance?

33. The manufacturer of surgical instruments claims that less than 1% of the instruments he supplied to a certain hospital are faulty. A sample of 300 instruments revealed that 6 were faulty. Test his claim at 1% level of significance.

34. In a random sample of 1000 persons from a large population, 470 are women. Can we conclude that men and women are in the equal ratio in the population? (Use $\alpha = 0.05$).

35. In an election the leaders of a party contend that they would secure more than 36% of votes. A pre-poll survey of 400 voters revealed that the percentage is 42. Does the survey support the leader's claim? (Use 1% L.O.S)

36. In a random sample of 100 II PUC Statistics students 9 secured distinctions. At 5% level of significance, can we conclude that 10% of II PUC Statistics students secured distinctions?

37. A stock broker claims that he can predict with 80% accuracy whether a stock market value will rise or fall during the coming month. In a sample of 40 predictions, 28 are correct. Does this evidence supports broker's claim at 1% level of significance?

38. The manufacturer of surgical instruments claims that less than 2% of the instruments he supplied to a certain hospital are faulty. A sample of 400 instruments revealed that 12 were faulty. Test his claim at 5% level of significance.

39. In a random sample of 80 persons from town A, 30 are found to be consumers of wheat. In a random sample of 40 persons from town B, 18 are found to be consumers of wheat. Does this data reveal a significant difference between proportion of wheat consumers in town A and town B? Use 5% L.O.S.

40. In an institution, out of 500 students who took S.S.L.C. examination, 460 passed and out of 400 students who took P.U.C. examination, 350 passed. At 1% level of significance, can it be concluded that S.S.L.C. students have performed better than P.U.C. students?

41. A machine produced 26 defective articles among 250. Another machine produced 4 defective articles among 50. Test whether there is any significant difference between population proportions. Use 5% level of significance.

42. Among 400 students randomly selected from college A, 72% of students passed. Among 200 students randomly selected from college B, 66% of students passed. Can it be concluded that performance of college A is better than performance of college B? Use 5% level of significance.

43. In a random sample of 120 people from a city in the year 2011 revealed that 96 were cricket match viewers. In another random sample of 100 people from same city in the year 2013 revealed that 90 were cricket match viewers. Examine whether there is a significant increase in the proportion of cricket match viewers. Use 1% level of significance.

44. Among randomly selected 100 students of college A, 66 were passed. Among randomly selected 200 students of college B, 144 were passed. Test whether passing proportion is same in both the colleges. Use 5% L.O.S.

45. Among 80 randomly selected persons from district A, 36 are interested in viewing hockey match. Among 40 randomly selected persons from district B, 12 are interested in viewing hockey match. Test at 5% level of significance that, the proportion of viewers in district A is more than district B.

46. From the following data test whether, the difference between population proportions is significant at 5% level of significance.

Sample	I	II
Size	200	100
Proportion	0.28	0.34

47. From the following data test whether, the difference between population proportions is significant at 1% level of significance.

Sample	I	II
Size	100	400
Proportion	0.02	0.01

48. From the following data test whether, the difference between population proportions is significant at 5% level of significance.

Sample	I	II
Size	200	100
Proportion	0.12	0.09

Unit VI (c)
t- Tests

One mark questions:

1. Write an application of t-test.

Two mark questions:

1. Write two applications of t-test.
2. Write down the t-test statistic and degrees of freedom in case of test for mean.
3. Given, $\bar{x} = 53$ gm, $\mu = 50$ gm, s = 5 gm and n = 17, calculate test statistic t.
4. Write down the t-test statistic and degrees of freedom in case of test for equality of means of two independent samples.
5. Write down the test statistic and degrees of freedom of paired t-test.
6. In paired t-test, if n = 5, $\bar{d} = 3$ and $s_d = 2$, then what would be the value of test statistic t?

Five mark questions:

1. The length of 10 samples of cotton taken from a population has mean length of 48cm and S.D. 3cm. Test whether the mean length of the population can be taken as 50cm? Use 5% level of significance.

2. It is required to test whether those who practice yoga have average blood sugar less than 120 mg/dl. A sample consisting of 17 persons who practice yoga is observed. If their mean blood sugar is 108 mg/dl and S.D. is 8 mg/dl. What would you conclude? Use 5% level of significance.

3. The mean weekly sales of the chocobar in candy stores were 146.3 bars. After advertising campaign the mean weekly sales in 22 stores for a typical week increased to 153.7 bars and showed S.D. of 17.2 bars. Was the advertisement campaign successful? Use $\alpha = 5\%$

4. A random sample of size 20 is taken from normal population gives a sample mean 42 and standard deviation 6. Test the hypothesis that the population mean is 44. Use $\alpha = 5\%$.

5. A company has been producing steel tubes with mean inner diameter of 2 cm. A sample of 10 tubes gives a mean inner diameter of 2.01 cm and a variance of 0.004 cm^2. Is the company meeting the specification? Use $\alpha = 5\%$

6. A manufacturer is making axle with diameters 0.7". A random sample of 10 axles shows a mean diameter of 0.742" with S.D of 0.04". Test whether manufacturer is meeting the specification. Use $\alpha = 5\%$.

7. The mean weekly sale of ice-cream bars was 140 bars. After an advertising campaign the mean weekly sale in 26 shops for a typical week increased to 150 and showed a S.D. 20. Is this evidence indicates that the advertising campaign successful? Use $\alpha = 1\%$.

8. A soap manufacturing company was distributing a particular brand of soap through a large number of retail shops. Before a heavy advertisement campaign, the mean sale per week per shop was 140 dozens. After the campaign a sample of 26 shops was taken and mean sales was found to be 147 dozens with standard deviation 16 dozens. Is advertisement effective? Use $\alpha = 5\%$.

9. A survey in a locality revealed that on an average 180 persons read a particular newspaper. But the newspaper agent felt that it is more than 180. He conducted a survey in 10 localities and it was found that on an average 185 persons read that newspaper with S.D 6. Conduct the test at 1% level of significance.

10. The weights of 5 students of a college are 45, 47, 48, 49 and 51kg. Test whether the average weight of the students of the college is 50kg. Use $\alpha = 5\%$

11. Five students are selected at random from a college and their heights are found to be 149, 151, 152, 153 and 155cm. Test at 5% level of significance that the average height of the students of the college is 150 cm.

12. A random sample of size 16 has mean 53. The sum of the squared deviations taken from mean is 150. Can this sample be regarded as taken from the population having mean 56? Use $\alpha = 0.01$.

13. Mean and S.D. of heights of persons of two localities gave the following results.

	Locality A	Locality B
Sample size	12	8
Mean (cm)	175.3	177.7
S.D.(cm)	4.2	3.7

Can we conclude at 5% L.O.S. that the population of locality A on an average is shorter than locality B?

14. For the following data test whether mean weight of group-I is greater than group-II (use k=1.81)

	Group-I	Group-II
Sample size	7	5
Mean (kg)	50	48
Variance(kg^2)	5	3

15. The marks obtained by two groups of students in a statistics test are given below:

	Group A	Group B
No. of students	15	11
Mean marks	42	38
S.D. marks	10	15

On the basis of above data, can it be concluded that there is a significant difference in the mean marks obtained by the two groups? Use $\alpha = 5\%$.

16. For the following data examine whether the means differ significantly? Use $\alpha = 5\%$. (U)

Sample	I	II
Size	12	7
Mean	57.2	52.3
S.D.	3.41	3.62

17. The average weight of 6 randomly selected women is 68 kg and that of 10 randomly selected men is 67.8 kg. Their variances of weights are 12 kg^2 and 17 kg^2 respectively. Test whether the average weight of women is more than men. Take $\alpha = 0.01$.

18. I.Q. of 5 students before and after training is given below:

Student	1	2	3	4	5
Before training	110	123	120	132	125
After training	120	125	118	136	121

Test whether the training improves the I.Q. of students. (Take $\alpha = 0.01$)

19. The following data represents the blood pressure of 5 persons before and after performing dhyana:

Person	A	B	C	D	E
B.P. Before Dhyana	90	90	100	88	99
B.P. After Dhyana	88	90	95	90	96

Can we conclude at 5% level of significance that Dhyana reduces blood pressure?

20. Following is the data regarding the I.Q. of five students before and after treatment of Yoga:

I.Q. Before	118	120	116	115	125
I.Q. After	125	118	125	120	130

Is Yoga improves the I.Q. of students? Use 5% level of significance.

21. An I.Q. test was conducted to 5 students before and after training and following data is obtained.

Student	1	2	3	4	5
I.Q. Before	121	128	120	118	125
I.Q. After	125	132	110	120	127

Test whether there is any significant difference between average I.Q. before and after training. (Take $\alpha = 0.01$).

22. Five students were given intensive coaching and tests were conducted before and after coaching. The scores of the tests are given below. Do the scores after coaching show an improvement? (Use $\alpha = 0.05$).

Student No	1	2	3	4	5
Marks before coaching	50	42	51	26	35
Marks after coaching	62	40	61	35	40

23. In geometry, first test was conducted to five boys of a school. They were given a month tuition and second test was held at the end of it. The scores of the tests are given below. Test whether the tuition was benefited the students. (Use $\alpha = 0.05$).

Boys	1	2	3	4	5
Marks (I Test)	23	20	19	21	18
Marks (II Test)	24	18	21	18	20

24. A certain drug administered to 10 patients resulted in the following change in blood pressure:-5, 2, -1, -3, 0, -2, 1, 5, 0, -4
Can it be concluded that the drug will decrease the blood pressure? Use $\alpha = 0.05$.

25. Ten hyper tension patients, to whom a certain drug was administered, change in blood pressure are as follows:-7, 3, -1, 4, -3, 5, -6, -4, 1, -2
Can it be concluded that the drug will decrease the blood pressure? Use $\alpha = 0.05$.

Unit VI (d)
Chi-square Tests

One mark questions:

1. Mention an application of χ^2- test.
2. Write the degrees of freedom in testing of population variance for a sample of size n.
3. For the χ^2- test, what is the condition for expected frequency?
4. When do you pool the expected frequencies in testing of goodness of fit?
5. What is the value of degrees of freedom in testing of independence of attributes in 2×2 contingency table?

Two mark questions:

1. Mention two applications of χ^2- test.
2. Write the χ^2 - test statistic with degrees of freedom in testing of population variance for a sample of size n.
3. Write the χ^2 - test statistic with degrees of freedom in testing of goodness of fit.
4. Mention two conditions for applying χ^2- test of goodness of fit.
5. In a χ^2- test for goodness of fit, if there are 6 observations in which one parameter is estimated, then find the value of degrees of freedom of test statistic.
6. Write the χ^2- test statistic with degrees of freedom in testing of independence of attributes in 2×2 contingency table.
7. Write two conditions for applying χ^2- test of independence of attribute.

Five mark questions:

1. A normal variate has variance 8. Twenty sample observations of the variate have variance 9. Test at 1% level of significance whether the population variance is 8.

2. A normal variate has variance 81. Twenty one random observations of the variate have variance 100. Test at 1% level of significance whether the sample variance differs significantly from the population variance.

3. The standard deviation of production of paddy is assumed to be 10.6 tons. A sample of 20 acres showed that the standard deviation is 8.3 tons. Test at 1% level of significance whether the standard deviation of production of paddy is less than 10.6 tons.

4. A random sample of size 25 taken from a population gives the sample standard deviation 8.5. Test the hypothesis that the population standard deviation (σ) is 10. Useα= 0.05.

5. A manufacturer claims that the life time of batteries produced by the factory has variance 4000 hours^2. To test this, a sample of 18 batteries were tested and found the variance of 6000 hours2. Test the manufacturer's claim at α= 0.05.

6. Test the hypothesis σ = 5, given that sample standard deviation is 6 for a random sample of size 25 from a normal population. Use α= 0.05.

7. The variance of the height of 20 SSLC students is 4 cm^2. Test at 1% level of significance that the variance of height of SSLC students is more than 3 cm^2.

8. Following are the points scored by five students in a competition:
   1, 13, 9, 5, 7
   Test at 5% level of significance that the population variance is more than 15.

9. The tensile strength of 8 rods were 8, 3, 12, 14, 7, 13, 9 and 6 tons. Test the hypothesis that the standard deviation is more than 2 tons. Use 5% L.O.S.

10. Weights in kg. of 10 students are given below:
    38, 40, 45, 53, 47, 43, 55, 48, 52, 49
    Can we say that variance of the distribution of weights is equal to 20 kg^2? Use 5% L.O.S.

11. Weights of 10 students are given below:
    Weight (in kg) : 32, 48, 50, 47, 49, 55, 46, 51, 46, 50.
    Can you conclude that standard deviation of the distribution of weights of students is 4 kg? Use 1% L.O.S.

12. Binomial distribution is fitted to an observed frequency distribution after estimating 'p' from the observed data. The observed and the expected frequencies are given below.

X	0	1	2	3	4	5
O_i	14	56	110	88	40	12
E_i	10	50	100	100	50	10

Test whether B.D. is a good fit. Use α = 0.05.

13. From the following data, test whether the Poisson distribution is a good fit. The values are tabulated after estimating the parameter. Use $\alpha = 0.01$.

O_i	46	29	12	7	4	2
E_i	37	37	18	6	2	0

14. 70 accidents that have occurred in a state in a week are tabulated as follows:

Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Accidents	7	8	11	12	5	13	14

Test whether accidents occur uniformly throughout the week. (k_2 = 16.81)

15. In 120 throws of a die, the following distribution of faces was obtained.

Faces	1	2	3	4	5	6	Total
Frequency	30	25	18	10	22	15	120

Test at 5% level of significance that the die is unbiased.

16. A die is thrown 120 times and the results are as follows.

Face of die	1	2	3	4	5	6
Frequency	12	18	20	32	20	18

At 5% level of significance, test whether the die is fair.

17. Among 64 offspring of certain cross breed of guinea pigs, 34 were red, 10 were black and 20 were white. According to genetic model, these numbers should be in the ratio of 9 : 3 : 4. Are the data consistent with model at the 5% level of significance?

18. A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured 3rd class, 90 were placed in 2nd class and 20 got 1st class. Do these figures commensurate with the general examination result which is in the ratio of 4 : 3 : 2 : 1 for various categories respectively.

19. The following is the data regarding family condition and examination result of 100 students. Test whether family conditions and results are independent. (k_2 = 6.63)

Family Condition	Examination Result	
	Pass	Fail
Good	30	10
Bad	20	40

20. From the following data, test whether 'education' and 'employment' are independent at 1% level of significance.

Education	Employment	
	Employed	Unemployed
Educated	20	25
Uneducated	15	40

21. From the following data test whether the attributes 'literacy' and 'smoking' are independent. Use 5% level of significance.

	Literates	Illiterates
Smokers	7	18
Non-smokers	13	12

22. An opinion poll was conducted to find the reaction to a proposed civic reform in 100 members of each of the two political parties. The information is tabulated below.

Party	Favourable	Unfavourable
A	40	60
B	50	50

Test whether political parties and the reaction to a proposed civic reform are independent.

23. From the following data test whether the examination result depends on special coaching. Use 1% level of significance.

Special coaching	Examination Result	
	Pass	Fail
Taken	210	90
Not taken	60	40

24. Consider the following 2×2 contingency table. Test whether X and Y are independent at 1% level of significance.

X / Y	X_1	X_2
Y_1	6	24
Y_1	14	16

25. 40 students of college A and 80 students of college B are appeared for an examination. The results are as follows.

College	Examination Result	
	Pass	Fail
A	26	14
B	60	20

Is result depends on college? Use $\alpha = 0.05$

26. In an experiment of immunization of cattle for tuberculosis, the following results were obtained.

Vaccine	Affected	Unaffected
Inoculated	12	26
Non-inoculated	16	6

Test whether the vaccine is effective in controlling tuberculosis. Use $\alpha = 0.01$.

27. Of the 500 workers in a factory exposed to an epidemic, 350 were attacked, 200 had been inoculated and of these 100 were attacked. Test whether inoculation and attack of epidemic are independent. Use $\alpha = 0.05$

28. In a survey of 200 boys 75 were intelligent. Among intelligent boys 40 had skilled fathers. While 85 ofthe unintelligent boys had unskilled fathers. Do these figures support the hypothesis that skilled father have intelligent boys? Use $\alpha = 0.05$

UNIT – VII STATISTICAL QUALITY CONTROL

Choose the most appropriate answer from the choices given.

1. Suppose X is a variable quality characteristic, which chart is used for the control of average quality level?
   a) $\overline{X}$ – chart b) R- chart c) np-chart d) c-chart

2. Suppose X is a variable quality characteristic, which chart is used for the control of variation?
   a) $\overline{X}$ – chart b) R- chart c) np-chart d) c-chart

3. Which control chart is used for number of defectives?
   a) $\overline{X}$ – chart b) R- chart c) np-chart d) c-chart

4. Which control chart is used for number of defects?
   a) $\overline{X}$ – chart b) R- chart c) np-chart d) c-chart

5. For R-chart if σ' is known, then the central line is
   a) σ' b) d_2 σ' c) D_1 σ' d) D_2 σ'

Fill in the blanks by choosing the appropriate word from those given in the brackets.

[Attribute, Assignable cause, R- chart, Chance cause, Product control, Defect, Process control, Defective, Variable, Control chart, np-chart, d_2σ', c-chart]

1. Statistical quality control helps in detecting _____type of variation.
2. The process of inspection of manufactured lot for acceptability is called_____.
3. A _____is a quality characteristic which does not conform to specifications.
4. An item having one or more defects is a _____item.
5. Controlling the quality of goods during the manufacturing process itself is called _____.
6. In statistical quality control, a measurable quality characteristic, which varies from unit to unit, is _____.
7. A qualitative characteristic which cannot be measured and can only be identified by its presence or absence is an _____
8. A small amount of variation for which no specific cause can be attributed is termed as _____of variation.
9. _____is a graphical device which is used to study and verify whether the production process is under statistical control or not.
10. In statistical quality control, $\overline{X}$ – chart and _____are used for the control of variables.
11. In statistical quality control, _____is used for number of defectives.
12. In statistical quality control, _____is used for number of defects.
13. For R-chart if σ' is known, then _____ is the central line.

Match the following (Direct answers are given)

	$\overline{X}$ – chart		Control of average quality level
	R-chart		Control of variation
	np-chart		Number of defectives
	c-chart		Number of defects
	Chance variation		Beyond human control
	$\overline{X}$ – chart & R-chart		Control charts for variables
	np-chart & c-chart		Control charts for attributes

One mark questions:

1. Define Statistical Quality Control.
2. Mention a type of cause for variation in a manufacturing process.
3. What are chance causes?
4. What are assignable causes?
5. Which type of 'cause of variation' is detected by SQC?
6. Which cause of variation cannot be detected by SQC?
7. Mention an advantage of SQC.
8. What do you mean by process control?
9. What do you mean by product control?
10. What is a defect in SQC?
11. Give an example for defect.
12. What are defectives in SQC?
13. Give an example for defectives.
14. What is a control chart?
15. What is upper control limit?
16. What is lower control limit?
17. Mention different types of control charts.
18. Name control charts for variables.
19. Name control charts for attributes.
20. Name the control chart used for number of defectives.
21. Name the control chart used for number of defects per unit.
22. In $\overline{X}$- chart, if one of the sample mean lies outside the control limits, what would you conclude?
23. In a control chart, if one point lies outside the control lines, what is your inference?
24. Define Acceptance sampling.
25. Write an advantage of 'Acceptance sampling'.
26. Write a disadvantage of 'Acceptance sampling'.

Two mark questions:

1. Mention two types of causes for variation in a manufacturing process.
2. What are chance and assignable causes?
3. Mention two advantages of SQC.
4. What do you mean by product control and process control?
5. In SQC define variable and attribute.
6. In SQC what is a defect? Give an example.

7. In SQC what are defectives? Give an example.
8. In SQC what are defect and defectives?
9. Give one example each for defect and defectives.
10. Name the control charts for variables and attribute.
11. Write the upper and lower control limits for $\bar{X}$ - chart, when standards are known.
12. Compute upper control limit for $\bar{X}$ - chart when $\bar{X}' = 40$, $\sigma' = 6$ and A = 1.342.
13. Write the upper and lower control limits for $\bar{X}$ - chart, when standards are not known.
14. If $\bar{\bar{X}} = 40$, $\bar{R} = 2.5$ and $A_2 = 0.577$, find lower control limit of $\bar{X}$ - chart.
15. Write the upper and lower control limits for R-chart when standards are known.
16. If $\sigma' = 4$ and n = 5, find average quality level [C.L.] of R-chart.
17. Write the upper and lower control limits for R-chart when standards are not known.
18. If $\bar{R} = 2.5$ and n = 4, find upper control limit of R-chart.
19. Write the upper and lower control limits for d-chart when standards are known.
20. If $P' = 0.1$ and n = 100, calculate U.C.L. for d-chart.
21. Write the upper and lower control limits for d-chart when standards are not known.
22. Write the upper and lower control limits for c-chart when standards are known.
23. Write the upper and lower control limits for c-chart when standards are not known.
24. In a floor mat manufacturing company, the average number of defects per square meter of floor mat is known to be 4. Find the upper control limit for the number of defects.
25. If $\bar{c} = 2$, find U.C.L. for c - chart.
26. Write two advantages of acceptance sampling.
27. Write relative merits and demerits of single sampling plan and double sampling plan.

Five mark questions:

1. A drilling machine is set to dig holes with a mean diameter of 15mm and a standard deviation of 0.2mm. Find the control limits for mean, for samples of size 5 each.

2. If $\bar{X}' = 4.5$, $\sigma^1 = 1.5$ for samples of size n = 4, compute control limits of $\bar{X}$ - chart.

3. If $\bar{\bar{X}} = 40$, $\bar{R} = 2.5$ for samples of size n = 5, compute control limits of $\bar{X}$ - chart.

4. The measurement of 8 sub groups of samples of size 4 each gave the following results: $\bar{\bar{X}} = 33.3$ and $\bar{R} = 8.2$. Write down the control limits for $\bar{X}$ -chart.

5. For the following data, find control limits of $\bar{X}$ -chart. (Given, $A_2 = 0.729$)

Sub-group	1	2	3	4	5	6	7	8	Total
Mean	52	48	53	49	50	48	53	47	400
Range	10	11	8	12	9	10	9	11	80

6. For the following data, find control limits of $\bar{x}$-chart. (Given, $A_2 = 0.729$)

Sub-group	1	2	3	4	5	6
Mean	52	49	53	48	51	47
Range	4	6	5	7	3	5

7. For the following data, find control limits of $\bar{x}$-chart. (Given, $A_2 = 0.577$)

Sub-group	1	2	3	4	5	6
Mean	49	52	53	51	47	48
Range	3	5	7	6	4	5

8. Given, n = 4 and R_i: 14, 8, 11, 9, 6 and 12. Find the control limits of R-chart.

9. Given, n = 5 and R_i: 6, 3, 8, 4, 1, 2, 5 and 7. Find the control limits of R-chart.

10. If $P^{|} = 0.02$ and n = 100, calculate the control limits of np -chart.

11. In a fish net manufacturing process, the proportion defectives $p^{|} = 0.01$. If the processis based on samples of size 100 each, find the control limits of np-chart.

12. Calculate the control limits for d-chart given $\bar{p} = 0.05$ and sample size 50.

13. In a mobile phone manufacturing plant, a production manager has inspected a sample of 25 phone sets at regular intervals. If the average fraction defective is $\bar{p} = 0.02$, write down the control limits for number of defectives.

14. Ten samples of 100 P.V.C. pipes manufactured by a firm are inspected for the number of defectives. The number of defective pipes is noted as: 2, 1, 3, 0, 2, 2, 4, 4, 5, 6. Calculate the control limits for np-chart.

15. A company manufactures flooring tiles. Samples of 100 tiles each are drawn at regular intervals. The number of defective tiles is given below.

Sample no.	1	2	3	4	5	6	7	8	9	10
No. of defective tiles	2	3	1	0	4	2	4	2	6	4

Obtain the control limits for the above data.

16. In a floor mat manufacturing company, the average number of defects per square meter is known to be 4. Determine the control limits for the number of defects.

17. In a printing industry at regular intervals, cloth is inspected for defects in printing. If on an average 0.5 defects are expected per square meter, obtain suitable control limits.

18. One meter of cloth was inspected for weaving defects. Total number of defects for 10 such samples was 24. Find the control limits for defects.

19. During an examination of equal length of cloth, the following are the number of defects observed as: 4, 5, 6, 6, 3, 2, 6, 7, 3, 4. Calculate control limits for suitable control chart.

20. Twenty pieces of cloth out of different rolls contained respectively 1, 4, 3, 2, 5, 4, 6, 7, 2, 3, 2, 5, 7, 6, 4, 5, 1, 2, 3, 8 imperfections. Find control limits for c - chart.

21. For the following data find the control limits for R- chart.
(Given, $D_3 = 0$, $D_4 = 2.115$ and $\bar{R} = 4$)

Sub-group	1	2	3	4	5	6	7	8
Range	4	5	5	3	6	2	4	3

22. In a textile mill, cloth is inspected at regular intervals for weaving defects. The results are recorded as below.

Sample number	1	2	3	4	5	6	7	8
Defects/sq. metre	2	1	0	0	2	5	1	1

If on an average 1 defect is expected per square meter, find the control limits for c-chart.

UNIT – VIII OPERATIONS RESEARCH

Choose the most appropriate answer from the choices given.

1. Graphical solution to the linear programming problem lies in the –
   a) I quadrant b) II quadrant
   c) III quadrant d) IV quadrant

2. Graphical solution to L.P.P. lies in the first quadrant because, -
   a) $x \geq 0, y \geq 0$ b) $x \leq 0, y \leq 0$ c) $x \geq 0, y \leq 0$ d) $x \leq 0, y \geq 0$

3. Transportation problem (T.P.) said to be balanced if and only if –
   a) $a_i = b_j$ b) $a_i < b_j$ c) $a_i > b_j$ d) $\Sigma a_i = \Sigma b_j$

4. In a T.P., the solution is said to be non-degenerate, when –
   a) The number of positive allocations in any B.F.S. > m+n–1
   b) The number of positive allocations in any B.F.S. < m+n–1
   c) The number of positive allocations in any B.F.S. = m+n–1
   d) The number of positive allocations in any B.F.S. ≠ m+n–1

5. In a transportation problem, any basic feasible solution is said to be non-degenerate when number of positive allocation is equal to
   a) m + n – 1 b) m – n + 1 c) n – m + 1 d) m + n + 1

6. A game is said to be fair, if value of the game is,
   a) v = 0 b) v ≠ 0 c) v = 1 d) v = 2

7. The maximum of row minimums is known as –
   a) Minimax b) Maximin c) Saddle point d) Strategy

8. The minimum of column maximums is known as –
   a) Minimax b) Maximin c) Saddle point d) Strategy

9. The problem of deciding the age at which old equipments are replaced by new ones is called –
   a) Linear programming problem b) Transportation problem
   c) Replacement problem d) Game problem

10. In replacement theory, the depreciation cost of the item is-
    a) $P - S_n$ b) $S_n - P$ c) $P + S_n$ d) ΣC_i

11. The cost associated with the maintenance of an inventory until they are sold or used is called:
    a) Capital coat b) Setup cost c) Shortage cost d) Holding cost.

12. The time gap between the placing of order and arrival of goods at inventory is called:
    a) Demand b) Time horizon c) Lead time d) Stock replenishment

13. The time period over which the inventory control is planned is called:
    a) Demand b) Time horizon c) Lead time d) Stock replenishment

Fill in the blanks by choosing the appropriate word from those given in the brackets.

[Transportation, Basic, $\Sigma a_i = \Sigma b_j$, Maximin, Game, Strategy, First quadrant, Zero, Less than, Minimax, Lead time, Setup, Feasible region, Equal to, Pay-off matrix, Saddle point, Holding cost, Time horizon, Value]

1. Graphical solution to L.P.P. lies in the_____.
2. _____is the area which satisfies all the constraints and also non-negativity restrictions.
3. _____problem helps in shipment of quantities from various sources to different destinations at a minimum cost.
4. A Transportation problem is balanced if and only if _____.
5. In a T.P., (m+n–1) variables are known as _____variables.
6. When the number of positive allocations in any B.F.S. is _____(m+n–1), the solution is said to be degenerate.
7. When the number of positive allocations in any B.F.S. is _____(m+n–1), the solution is said to be non-degenerate.
8. _____theory helps in deciding the best strategy in conflicting situations.
9. The _____of a player is the pre-determined rule by which a player determines his course of action.
10. A game is said to be fair if the value of the game is_____.
11. _____is the table showing how payments are to be made at the end of each game.
12. The maximum of the row minimums is known as _____.
13. The minimum of the column maximums are known as _____.
14. The position of the pay-off matrix where maximin and minimax coincide is the _____.
15. The pay off at the position of the saddle point is called _____of a game.
16. The cost associated with the maintenance of an inventory until they are sold or used is called _____.
17. The time gap between the placing of order and arrival of goods at inventory is the _____.
18. The time period over which the inventory control is planned is called the _____.
19. Inventory reduces the _____cost.

Match the following. (Direct answers are given)

	Graphical solution to L.P.P.		First quadrant
	$\Sigma a_i = \Sigma b_j$		Balanced T.P.
	$\Sigma a_i \neq \Sigma b_j$		Unbalanced T.P.
	North-west corner rule method		Transportation problem
	Matrix minima method (Least cost entry method)		Transportation problem
	Value of the game, v = 0		Fair game
	Maximin		Maximum of row minimums
	Minimax		Minimum of column maximums
	Dominance principle		Game theory
	Depreciation cost		$P - S_n$
	Physical stock of goods		Inventory
	Holding cost		C_1
	Shortage cost		C_2
	Setup cost		C_3
	Profit due to increasing market price		Advantage of inventory
	Inventory Model – I		Shortages are not allowed
	Inventory Model – II		Shortages are allowed

One mark questions:

1. Write a scope of Operations Research.

Two mark questions:

1. Define Operations Research.
2. Write two scope of operations research.
3. Mention two areas where the economic operations research can be applied.

(a) Linear Programming Problem (L.P.P)

One mark questions:

1. Define linear programming problem.
2. In L.P.P define decision variables.
3. In L.P.P define objective function.
4. In L.P.P define solution.
5. In L.P.P define feasible solution.
6. In L.P.P define optimal solution.
7. In L.P.P is an optimal solution always a feasible solution?
8. Mention a method of solving Linear Programming Problem.
9. When an L.P.P is said to have unique solution?
10. When an L.P.P is said to have multiple solutions?
11. When an L.P.P is said to have unbounded solution?
12. When an L.P.P is said to have no solution?
13. In graphical method why feasible solution of L.P.P lies in the first quadrant?

Two mark questions:

1. Give the general form of L.P.P. in matrix notation.
2. Mention two methods of solving Linear Programming Problem.
3. In which quadrant the feasible solution of the L.P.P lies in graphical method? Give reason.

4. In an L.P.P The objective function is Max Z = 20x + 70y and if feasible solutions are A (0, 8) and B(12 , 5). Find optimal solution.
5. The objective function and two feasible solutions of an L.P.P are Max Z = 200x + 100y and A(0,18), B(12,0). Find the optimal solution.
6. Which of the two feasible solutions (12, 10) and (14, 4) of an L.P.P maximizes the objective function Z = 5x + 4y?
7. Which of the two feasible solutions (10, 12) and (14, 8) of an L.P.P Minimizes the objective function. Min. Z = 10x + 20y?
8. Consider the LPP: Max. $Z = 3x + 5y$

   s.t. $x + 2y \leq 9$

   $x \leq 3$

   and $x, y \geq 0$

   If x = -1 and y = 5 is a solution to L.P.P. Is it a feasible solution? Give reason.

Five mark questions:

1. A resourceful home decorator manufactures two types of lamps A and B. Both the lamps go through two technicians, a cutter and a finisher. Lamp A requires two hours of cutter's time and one hour of finisher's time. Lamp B requires one hour of cutter's time and two hours of finisher's time. The cutterhas 104 hours and finisher has 76 hours of available time per month. Profit on one lamp of A variety is Rs.10 and B variety is Rs.12. Formulate the L.P.P.

2. A firm can produce two types of cloth say, A and B. Three kinds of wool are required for it, red, green and blue. One unit length of type A cloth needs 2 yards of red wool and 3 yards of blue wool; one unit length of type B cloth needs 3 yards of red wool, 2 yards of green wool and 1 yard of blue wool. If the firm has a stock of 15 yards of red wool, 8 yards of green wool and 12 yards of blue wool. Profit from unit length of cloth A and B is Rs. 5 and Rs. 8 respectively. Formulate the L.P.P.

3. A small manufacturer employs 5 skilled men and 10 semi-skilled men for making a product in two qualities, a deluxe model and an ordinary model. The production of a deluxe model requires 2 hours of skilled man and 3 hours of semi-skilled man. The ordinary model requires 1 hour of skilled man and 2 hours of semi skilled man. According to worker union rules, no man can work more than 8 hours per day. The profit from each deluxe model is Rs.10 and that from each ordinary model is Rs.8. Formulate a L.P.P such that the total profit is maximised.

4. A tailor gets a profit of Rs.100 from a shirt and Rs.170 from a pant. In a week from available 56 hours, he uses 36 hours for cutting and 20 hours for stitching. For cutting he requires 2 hours for a shirt and 3 hours for a pant. He requires 1 hour for stitching a shirt and 2 hours for stitching a pant. Formulate the L.P.P.

5. A manufacturer produces 2 products , A and B which needs two machines P and Q. Product A requires 6 hours on machine P and 2 hours on machine Q. Product B requires 4 hours on machine P and 4 hours on machine Q. There are 60 hours of time available on machine P and 80 hours on machine Q. Profit earned by the manufacturer on selling one unit of product A is Rs. 20 and one unit of product B is Rs. 12. Formulate the L.P.P.

6. Solve the following L.P.P graphically:
   Max. $Z = 40x + 20y$
   s.t. $2x + 3y \leq 12$
   $x + y \geq 3$
   and $x, y \geq 0$

   OR

   (For visually challenged students only)
   Write the procedure of solving an L.P.P by graphical method.

7. Solve the following L.P.P graphically:
   Min. $Z = 10x + 5y$
   s.t. $4x + 2y \leq 16$
   $2x + 3y \geq 12$
   and $x, y \geq 0$

   OR

   (For visually challenged students only)
   Write down the steps in the graphical method of solving L.P.P.

8. Solve the following L.P.P graphically:
   Max. $Z = 10x + 15y$
   s.t. $x + y \geq 10$
   $3x + 2y \leq 60$
   and $x, y \geq 0$

   OR

   (For visually challenged students only)
   Explain graphical method of solving L.P.P.

9. Solve the following L.P.P graphically:
   Max. $Z = 3x+5y$
   s.t. $x+2y \leq 20$
   $y \leq 6$
   and $x,y \geq 0$

   OR

   (For visually challenged students only)
   Write the procedure of solving an L.P.P by graphical method.

10. Solve the following L.P.P graphically:
    Min. $Z = 5x + 4y$
    s.t. $4x + y \geq 40$
    $2x + 3y \geq 60$
    and $x , y \geq 0$

    OR

    (For visually challenged students only)
    What is an L.P.P? Explain different kinds of solutions.

11. Solve the following L.P.P graphically:
Max. $Z = 3x+5y$
s.t. $x + y \leq 150$
$y \leq 60$
and $x, y \geq 0$

OR

(For visually challenged students only)
Explain graphical method of solving L.P.P.

12. Solve the following L.P.P graphically:
Minimize $Z = 10x + 5y$
Subject to $2x + 3y \geq 12$
$4x + 2y \geq 16$
and $x \geq 0, y \geq 0$

OR

(For visually challenged students only)
Write the procedure of solving an L.P.P by graphical method.

13. Solve the following L.P.P graphically:
Maximize $Z = 100x + 170y$
Subject to $2x + 3y \leq 36$
$x + 2y \leq 20$
and $x \geq 0, y \geq 0$

OR

(For visually challenged students only)
A tailor gets a profit of Rs.100 from a shirt and Rs. 170 from a pant. In a week from available 56 hours, he uses 36 hours for cutting and 20 hours for stitching. For cutting he requires 2 hours for a shirt and 3 hours for a pant. He requires 1 hour for stitching a shirt and 2 hours for stitching a pant. Formulate the L.P.P.

14. Solve the following L.P.P graphically:
Maximize $Z = 20x + 12y$
Subject to $6x + 2y \leq 60$
$4x + 4y \leq 80$
and $x \geq 0, y \geq 0$

OR

(For visually challenged students only)
A manufacturer produces 2 products , A and B which needs two machines P and Q. Product A requires 6 hours on machine P and 2 hours on machine Q. Product B requires 4 hours on machine P and 4 hours on machine Q. There are 60 hours of time available on machine P and 80 hours on machine Q. Profit earned by the manufacturer on selling one unit of product A is Rs. 20 and one unit of product B is Rs. 12. Formulate the L.P.P.

(b) Transportation Problem (T.P)

One mark questions:

1. In a T.P, define a feasible solution.
2. Define basic feasible solution (BFS) of a T.P.
3. When do you say that a basic feasible solution of T.P is degenerate?
4. When do you say that a basic feasible solution of T.P is non-degenerate?
5. Define optimal solution of a T.P.
6. When T.P is said to be balanced?
7. When a Transportation Problem said to be unbalanced?
8. Write the formula of total cost of T.P.
9. Mention a method of obtaining initial basic feasible solution to a T.P.

Two mark questions:

1. Define a transportation problem.
2. What do you mean by degenerate and non-degenerate solutions in T.P?
3. Mention two methods of obtaining initial basic feasible solution for a T.P.
4. Whether the following Transportation problem is balanced?

	D_1	D_2	D_3	Availability
O_1	2	5	2	180
O_2	8	3	7	100
O_3	6	2	10	120
Requirement	150	200	250	

5. Verify whether the following solution to a T.P is non-degenerate.

	D_1	D_2	D_3
O_1	30	10	
O_2		30	
O_3			20

6. Verify whether the following solution to a T.P is non-degenerate.

--	--	30	70
65	--	--	100
--	50	80	---
--	--	30	70

Five mark questions:

1. Determine an initial basic feasible solution to the following transportation problem by NWCR. Compute the transportation cost.

		To			Supply
		D_1	D_2	D_3	
From	O_1	2	7	4	5
	O_2	3	3	1	8
	O_3	5	4	7	7
	O_4	1	6	2	14
	Demand	7	9	18	

2. Determine an initial basic feasible solution to the following transportation problem by NWCR. Compute the transportation cost.

		To			Supply
		D_1	D_2	D_3	
From	O_1	8	4	12	500
	O_2	10	5	6	200
	O_3	7	5	3	100
	Demand	400	200	200	

3. Obtain an initial B.F.S for the following T.P by NWCR. Find the transportation cost.

		To			Supply
		X	Y	Z	
From	A	10	11	2	27
	B	8	9	6	33
	C	1	7	5	38
	D	3	14	12	22
	Demand	46	44	30	

4. For the following T.P, obtain an initial B.F.S by NWCR and show that it is degenerate.

		Destination				a_i
		A	B	C	D	
Origin	I	3	4	2	6	35
	II	4	3	3	2	50
	III	6	4	7	9	15
	b_j	25	45	15	15	

5. Determine an initial basic feasible solution to the following transportation problem by NWCR. Compute the transportation cost. Is the solution degenerate?

		To			Availability
		X	Y	Z	
From	A	8	7	3	60
	B	3	8	9	70
	C	11	3	5	80
	Requirement	50	80	80	210

6. Obtain an initial basic feasible solution to the following T.P by NWCR. Compute the transportation cost. Is the solution to T.P degenerate?

		Warehouse				Availability
		D_1	D_2	D_3	D_4	
Factory	O_1	19	30	50	10	70
	O_2	70	30	40	60	90
	O_3	40	8	70	20	180
	Requirement	70	80	50	140	

7. Find an initial basic feasible solution by Matrix Minima Method and compute the total transportation cost. Is the solution to T.P non degenerate?

		To			Supply
		D_1	D_2	D_3	
From	O_1	8	4	12	500
	O_2	10	5	6	200
	O_3	7	5	3	100
	Demand	400	200	200	

8. Determine an initial basic feasible solution to the following transportation problem by Matrix Minima Method and compute the transportation cost.

		To			Supply
		D_1	D_2	D_3	
From	O_1	2	7	4	5
	O_2	3	3	1	8
	O_3	5	4	7	7
	O_4	1	6	2	14
	Demand	7	9	18	

9. For the following transportation problem, find the initial basic feasible solution by Matrix Minima Method and obtain the transportation cost.

		Ware house			Availability
		I	II	III	
Factory	A	15	10	9	350
	B	5	8	9	100
	C	10	6	4	110
	Requirement	80	150	330	

10. Obtain an initial basic feasible solution to the following T.P by Matrix Minima Method. Compute the transportation cost.

		To			Supply
		A	B	C	
From	I	7	3	4	2
	II	2	1	3	3
	III	3	4	6	5
	Demand	4	1	5	

11. For the following T.P, find an initial B.F.S by Matrix Minima Method. Obtain the transportation cost.

		Destination				Availability
		D_1	D_2	D_3	D_4	
Origin	O_1	15	14	18	15	300
	O_2	17	19	15	10	250
	O_3	21	25	14	11	150
	Requirement	100	200	150	250	

(c) Game Theory

One mark questions:

1. Write a property of a competitive game.
2. What is n-person game?
3. What is two-person game?
4. What do you mean by zero-sum game?
5. Define rectangular game.
6. Define strategy in a game.
7. What do you mean by pure strategy in a game?
8. What do you mean by a mixed strategy in a game?
9. In a rectangular game, what is meant by pay-off matrix?
10. What is meant by maximin of a game?
11. What is meant by minimax of a game?
12. What is a saddle point?
13. When do you say that a game has saddle point?
14. What do you mean by the value of a game?
15. When the game is said to be fair?
16. If value of the game is -6, is the game fair?
17. In a rectangular game, if gain of a player is Rs.3 then what is the loss of the other?
18. If in a game the pay-off at saddle point is 4, what is the value of minimax?
19. Mention a method of solving a rectangular game.

Two mark questions:

1. Write two properties of a competitive game.
2. What are pure and mixed strategies in a game?
3. Explain maximin and minimax of a game.
4. What is a saddle point? When do you say that a game has saddle point?
5. What do you mean by the value of a game? When the game is fair?
6. Mention two methods of solving a rectangular game.
7. In a rectangular game the pay-off matrix of player A is given below. Write down the pay-off matrix of player B.

$$\text{Player B}$$
$$\begin{array}{cc} & \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \text{Player A} \begin{array}{c} A_1 \\ A_2 \end{array} & \begin{pmatrix} 5 & 3 & 8 & -6 \\ 5 & 10 & -3 & -6 \end{pmatrix} \end{array}$$

8. For the following pay-off matrix of player B, write down the pay-off matrix of player A.

$$\text{Player A}$$
$$\begin{array}{cc} & \begin{array}{ccc} A_1 & A_2 & A_3 \end{array} \\ \text{Player B} \begin{array}{c} B_1 \\ B_2 \end{array} & \begin{pmatrix} 1 & -3 & 5 \\ 2 & -4 & 6 \end{pmatrix} \end{array}$$

9. The following is the pay off matrix of player A. Write the pay-off matrix of player B.

$$\text{Player B}$$
$$\begin{array}{cc} & \begin{array}{cc} B_1 & B_2 \end{array} \\ \text{Player A} \begin{array}{c} A_1 \\ A_2 \\ A_3 \end{array} & \begin{pmatrix} 3 & 2 \\ 5 & 4 \\ 0 & -1 \end{pmatrix} \end{array}$$

10. Using maximin-minimax principle, find the value of game.

$$\text{Player A}\begin{array}{c} \\ A_1 \\ A_2 \end{array}\begin{array}{c} \text{Player B} \\ \begin{array}{cc} B_1 & B_2 \end{array} \\ \begin{pmatrix} 8 & 5 \\ 3 & 2 \end{pmatrix} \end{array}$$

Five mark questions:

1. Two players A and B play a game of tossing coins. If the coins match, then A gets Rs.5 from B and if the coins do not match, B gets Rs.10 from A. Write down the pay off matrix of A. Does the game have saddle point?

2. Players A and B play a game in which each player has 3 coins (Re.1, Rs.5, Rs.10). Each of them selects a coin without the knowledge of the other. If the sum of values of the coins is an even number, A wins B's coin otherwise B wins A's coin. Write down the pay off matrix of A. Does the game have saddle point?

3. For the following pay-off matrix, find the solution using maximin-minimax principle.

$$\text{Player A}\begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \end{array}\begin{array}{c} \text{Player B} \\ \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \begin{pmatrix} 10 & 8 & 6 & 20 \\ 20 & 12 & 7 & 9 \\ 3 & 2 & 3 & 2 \end{pmatrix} \end{array}$$

4. Solve the following game using maximin-minimax principle. Is the game fair?

$$\text{Company-X}\begin{array}{c} \\ A \\ B \\ C \\ D \end{array}\begin{array}{c} \text{Company-Y} \\ \begin{array}{ccc} P & Q & R \end{array} \\ \begin{pmatrix} 3 & -1 & 3 \\ 2 & -1 & 2 \\ -1 & 0 & 0 \\ 2 & 0 & 4 \end{pmatrix} \end{array}$$

5. Pay-off matrix of player A is as follows. Find the solution by maximin-minimax principle.

$$\text{Player A}\begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \\ A_4 \end{array}\begin{array}{c} \text{Player B} \\ \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \begin{pmatrix} -7 & 0 & 3 & -5 \\ 7 & -2 & 0 & -5 \\ -2 & -1 & -2 & 0 \\ 4 & 2 & 3 & 6 \end{pmatrix} \end{array}$$

6. Solve the following game using maximin-minimax principle.

$$\text{Player B}$$

$$\text{Player A}\quad \begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \end{array} \begin{array}{c} \begin{array}{ccc} B_1 & B_2 & B_3 \end{array} \\ \begin{pmatrix} 1 & 3 & 1 \\ 0 & -1 & -3 \\ 1 & 5 & -1 \end{pmatrix} \end{array}$$

7. Solve the following game using dominance principle.

$$\text{Player B}$$

$$\text{Player A}\quad \begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \end{array} \begin{array}{c} \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \begin{pmatrix} 5 & 2 & 1 & 6 \\ 2 & 1 & 0 & 2 \\ 7 & 5 & 4 & 5 \end{pmatrix} \end{array}$$

8. For the following pay-off matrix, find the solution using dominance principle. Is the game fair?

$$\text{Player B}$$

$$\text{Player A}\quad \begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \end{array} \begin{array}{c} \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \begin{pmatrix} 4 & 2 & 0 & 5 \\ -1 & -2 & 0 & -3 \\ -3 & 1 & -3 & 0 \end{pmatrix} \end{array}$$

9. Solve the following game using the principle of dominance.

$$\text{Player B}$$

$$\text{Player A}\quad \begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \end{array} \begin{array}{c} \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \begin{pmatrix} 1 & 2 & 0 & 3 \\ 4 & 6 & 3 & 5 \\ 3 & -1 & -2 & 0 \end{pmatrix} \end{array}$$

10. Solve the following game by dominance principle.

$$\text{Player B}$$

$$\text{Player A}\quad \begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \\ A_4 \end{array} \begin{array}{c} \begin{array}{ccc} B_1 & B_2 & B_3 \end{array} \\ \begin{pmatrix} 6 & 12 & 7 \\ 7 & 9 & 8 \\ 5 & 8 & 9 \\ 3 & 6 & 10 \end{pmatrix} \end{array}$$

(d) Replacement Theory

One mark questions:

1. What do you mean by replacement theory?
2. Mention a need for replacement of equipment.
3. When do you suggest for replacement of equipment which deteriorates with age?
4. Write the formula of to-date average annual cost.

Two mark questions:

1. Mention two needs for replacement of equipments.
2. If the depreciation cost and the cumulative maintenance cost of an equipment for the third year are Rs.10,000 and Rs.10,400 respectively. What is the average annual cost?
3. If the depreciation cost and the cumulative maintenance cost of an equipment for the second year are Rs.10,000 and Rs.10,200 respectively. What is the average annual cost?
4. For an equipment the fourth year depreciation cost is Rs.6,000 and the cumulative maintenance cost is Rs.6,200. Find the average annual cost.

Five mark questions:

1. The cost of a scooter is Rs 36,000. Its maintenance cost and resale value at different age are given below:

Year	1	2	3	4	5	6
Maintenance cost (Rs.)	800	1,300	1,900	2,700	3,900	5,400
Resale value (Rs.)	28,000	22,000	20,000	18,000	17,000	16,000

Determine the optimal age for replacement of the scooter.

2. The purchase price of a machine is Rs 10,000. Its maintenance costs and resale values are as below:

Year	1	2	3	4	5
Maintenance cost (Rs.)	600	800	1,000	1,400	2,000
Resale value (Rs.)	5,600	4,000	3,000	2,000	1,000

What would be the optimum replacement period of machine? What would be the average annual cost?

3. The purchase price of a machine is Rs 8,000. Its maintenance costs and resale values are given below:

Year	1	2	3	4	5
Maintenance cost (Rs.)	500	600	800	1,100	1,500
Resale value (Rs.)	4,500	3,500	2,500	1,500	500

What would be the optimum replacement period of machine? What would be the average annual cost?

4. Purchase cost of an item is Rs.5,000. Its running costs and resale values in different years are as follows:

Year	1	2	3	4	5
Running cost (Rs.)	100	200	330	510	860
Resale value (Rs.)	3,000	2,500	2,000	1,500	1,000

Find the optimum replacement period of item.

5. The following are the maintenance and depreciation costs per year of a vehicle.

Year	1	2	3	4	5	6	7
Depreciation cost (Rs.)	18,000	33,000	40,500	44,250	46,000	46,000	46,000
Maintenance cost (Rs.)	4,500	5,500	6,500	8,500	11,000	15,500	17,500

When the vehicle should be replaced?

6. The following are the maintenance and depreciation costs per year of a vehicle.

Year	1	2	3	4	5	6	7
Depreciation cost (Rs.)	20,000	35,000	42,500	46,250	48,000	48,000	48,000
Maintenance cost (Rs.)	5,000	6,000	7,000	9,000	11,500	16,000	18,500

When the vehicle should be replaced?

7. The cost of a machine is Rs.6600 and its resale value is Rs.600. The maintenance costs in different years are as follows:

Year	1	2	3	4	5	6	7
Maintenance cost (Rs.)	250	300	450	600	900	1500	1800

Find the annual average cost. What is the optimum period of replacement?

8. Price of an item is Rs. 1000 and its maintenance costs at different years are as follows:

Year	1	2	3	4	5
Maintenance cost (Rs.)	100	200	350	650	900

Assuming that resale value is negligible, find the annual average cost. What is the optimum period of replacement?

9. A machine costs Rs.36000 and the operating cost is estimated to be Rs.1500 for the first year and increases by Rs.3000 every year for next 5 years. Determine the optimum period for replacement of the machine, assuming that the machine has no resale value.

10. Equipment costs Rs. 5000, the running cost is Rs. 500 for the first two years and increases by Rs. 2000 from third year onwards. The scrap cost of the machine at all times is Rs. 300. Find the optimal replacement period.

11. Machine is priced Rs.6000 and operating cost is Rs.800 each for the first five years, increases by Rs.200 per year in the sixth and subsequent years. Determine the best age to replace the machine. Assume that the machine has no resale value.

(e) Inventory Theory

One mark questions:

1. What do you mean by inventory?
2. Mention one objective of inventory.
3. In inventory, what is holding cost?
4. In inventory, what do you mean by set-up cost?
5. In inventory, what do you mean by ordering cost?
6. In inventory, what do you mean by shortage cost?
7. In inventory, what is capital cost?
8. Mention a type of variable associated with inventory.
9. What is meant by stock replenishment in inventory?
10. Define uniform demand in inventory.
11. In inventory, what is lead time?
12. What is E.O.Q?
13. What is E.L.S?
14. Mention an advantage of inventory.
15. Write a disadvantage of inventory.
16. Mention a category of inventory model.

Two mark questions:

1. Mention two types of costs associated with inventory.
2. Mention two advantages of inventory.
3. State the assumptions of E.O.Q Model –I.
4. State the assumptions of E.O.Q Model –II.
5. Given, R = 3600 units/year, C_3 = Rs.50/cycle and C_1 = Rs.4/unit/year, find Q_0.
6. Calculate E.O.Q when R=5000 units/year, C_1=Rs.2/unit/year, C_3=Rs.200/cycle.
7. If R=100 units/month, C_3=Rs.250/cycle and C_1=Rs.0.20/unit/month. Find Q_0.
8. Given, R=5000items/year, C_3=Rs.50/cycle, C_1=Rs.2/item/year. Calculate minimum average inventory cost.

Five mark questions:

1. Given, Demand = 5000 items/year, Holding cost= Rs.2/item/year, Setup cost = Rs.50/cycle. Calculate i) E.O.Q ii) minimum average inventory cost.

2. There is a demand for 3600 units/year. The cost of placing an order is Rs. 50. Maintenance cost is Rs.9/unit/year. Find (i) E.O.Q (ii) Minimum average inventory cost.

3. If R = 12000 units/year, C_1 = Rs.0.3/unit/month, C_3 = Rs.150/run. Find (i) optimum lot size (ii) minimum average annual inventory.

4. There is a demand for 10000 items per year. The replenishment cost is Rs.200 and the maintenance cost is Rs.9 per item per year. Replenishment is instantaneous and shortages are not allowed. Find economic order quantity and minimum average inventory cost.

5. A manufacturer has to supply 12,000 units of a product per year to a customer. The inventory holding cost is Rs. 2 per unit per year and the set up cost per run is Rs. 750. Determine (i) the optimum lot size (ii) re-order time.

6. A stockist has to supply 300 units of a product every month to his customers. He gets the product at Rs.50 per unit from the manufacturer. The cost of carrying inventory is 12% per year of the cost of the product. The ordering cost is Rs.75/- per order. Find (i) EOQ and(ii) re-order frequency.

7. The annual demand for an item is 3000 units. Capital cost is Rs. 10 per unit. Inventory carrying cost is 20% of capital cost per annum. If setup cost is Rs.75, Find (i) EOQ (ii) re-order time.

8. Maruthi Udyog Company purchases 10,000 rear mirrors for cars annually. The ordering cost per order is Rs. 12. Each mirror costs Rs. 50 and the annual inventory carrying cost is 12% of capital cost. Compute EOQ and the minimum average inventory cost.

9. The demand for a commodity is at a constant rate of 200 units per year. There is an inventory in which setup cost is Rs. 800 per production run, holding cost is Rs.10 per unit per year and shortage cost is Rs. 12 per unit per year. Find the economic order quantity and maximum shortage level.

10. The demand for an item is 700 units per year. The cost of placing an order is Rs.7 and holding cost is Rs. 2 per unit per year. The shortage cost is Rs. 3 per unit per year. Find (i) EOQ (ii) maximum inventory level.

11. The following data gives various costs and other factors for the production of inventory systems of gears:

    Demand = 10000 gears / year
    Setup cost = Rs.180 / setup
    Holding cost = Rs. 40 / gear / year
    Shortage cost = Rs.100 / gear / year
    Find minimum average annual inventory cost

12. The annual demand of an item is 3600 units. The cost of placing an order is Rs.50 and annual maintenance cost is Rs. 9 per unit. The annual shortage cost is Rs. 16 per unit. Find (i) EOQ (ii) maximum inventory level.

13. The annual demand of an item is 8100 units. The cost of placing an order is Rs. 150 and annual maintenance cost is Rs. 3 per unit. The annual shortage cost is Rs. 9 per unit. Find (i) EOQ (ii) maximum shortage level.